Bachelor Thesis

A Model-based Computational Tool for Hybrid Rocket Engine Design

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Abstract

The primary aim of this thesis is to develop Computational Design Tool (CoDeT), a versatile and accurate tool for the preliminary design and performance prediction of hybrid rocket engines (HREs). CoDeT utilizes user specifications to determine crucial design parameters that optimize performance and subsequently models the combustion processes based on the regression rate law. Additionally, the program incorporates three flow models for injector design, suitable for common oxidizer types used in HRE systems.

CoDeT is specifically designed to cater to the needs of small to medium-scale hybrid propulsion systems throughout their development process. By utilizing the program, research groups and universities can reduce the need for extensive engine tests and computationally expensive simulations. The tool has the potential to significantly streamline the development process and improve performance predictions for future research and development endeavors.

To showcase the capabilities of the tool, a comprehensive case study is carried out, focusing on the design of a medium-scale HRE. The study involves a comparative analysis of three engines with similar specifications each utilizing a different oxidizer. On the basis of CoDeT results, respective performance and system efficiency of each engine are thoroughly evaluated, providing valuable insights into the decision-making process during the development of hybrid propulsion systems.

The results obtained from CoDeT compared with those generated by a commercially available development tool and corroborated with findings reported in existing research papers. The validation process indicates an overall good agreement. However, to conclusively determine the accuracy of the program, firing and cold-flow tests need to be conducted.

Keywords: Hybrid rocket engine, preliminary design tool, regression rate modeling of paraffin-wax

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Acronyms

CoDeT: Computational Design Tool

HRE: Hybrid Rocket Engine

HTPB: Hydroxyl-terminated polybutadiene

HTPE: High Density Polyethylene

O/F ratio: oxidizer-to-fuel ratio

AR: Expansion ratio or area ratio

IAC: Infinite Area Combustion

FAC: Finite Area Combustion

SH: Shower head

v.c.: vena contracta

CEA: Chemical Equilibrium with Applications

2 Introduction

A surge of interest in Hybrid Rocket Engines (HREs) in the last two decades [1] has triggered a need for preliminary design and performance prediction tools. Despite the highly accurate results provided by CFD simulations, the significant computational power demand and prolonged waiting times often render their utilization cumbersome during the initial stages of the design process. CFD software can accurately simulate subsystems such as the combustion chamber and nozzle; however, it lacks the capability to simulate the overall operation of the propulsion system [2]. Even though engine tests are a crucial part in the development of a HRE, accurate tools and software can reduce the number of required experiments and costly measuring capabilities. Considering the inherent limitations in accessing test facilities and the financial constraints faced by research groups and universities, the incorporation of computational tools become imperative. The primary aim of this thesis is to develop a versatile computational tool capable of both the preliminary design and predictive performance analysis of HREs. The developed tool, named **Co**mputational **De**sign **T**ool (CoDeT) is implemented in MATLAB and can be used by small research teams, universities, student rocket teams and hobby rocketeers.

The capabilities of CoDeT is showcased by presenting a case study involving a 1kN engine operating at 30 bar chamber pressure, with performance predictions conducted using three distinct oxidizers. Based on the results of CoDeT, hybrid propulsion system design parameters are investigated and implications to the real-world application of the designed engines are discussed in detail. The results obtained from CoDeT are compared with those generated by another development tool, namely Rocket Propulsion Analysis (RPA) [3], and corroborated with findings reported in existing research papers.

3 Literature Review and Theory

3.1 Overview of Hybrid Rocket Engines

HREs are a particular type of rocket engine that store propellants in diverse states. HREs pose important advantages over conventional solid and liquid rocket engines. These types of engines are considered safer and cheaper to manufacture and to transport due to the nature of the fuel types utilized. Moreover, the propellants being stored separately prevent an explosion caused by the intimate mixture of fuel and oxidizer [4]. The oxidizer and fuel types that are utilized by HREs also ensure the environmental safety of both the propellants and their combustion. Moreover, HREs offer flexibility in mission design. They allow for the adjustment of thrust value and vector steering, as well as multiple firing and long-term space operation. Further advantages of HREs include high operational stability, no abnormal combustion, and thus higher reliability [5].

HREs typically use a combination of liquid or gaseous oxidizer and inert solid fuel called *grain*. A wide range of propellant options are available, and the choice is typically dictated by mission requirements. For oxidizers, various options exist that offer differing levels of performance, including liquid and gaseous oxygen, liquid nitrous oxide, and hydrogen peroxide [6]. With the exception of natural rubbers, the most extensive group of viable fuels in terms of performance and mechanical characteristics is comprised of polymeric synthetic rubbers that originate from the polybutadiene monomer. Among these, the most prevalent option in terms of commercial availability and cost is hydroxyl-terminated polybutadiene (HTPB) [4].

The incorporation of performance additives, such as metal particles, can enhance the performance of HREs. Micron or nano-sized Al, B, and Mg or metal hydrides are some of the commonly used additives. Metallic additives increase the heat of combustion and the adiabatic flame temperature, resulting in an increased gas phase temperature and radiating particles, which ultimately lead to a higher burning rate. In addition, the use of these additives can improve the combustion product temperature, thus, enhancing the rocket motor's performance [7].

Despite the numerous advantages that HREs have over solid and liquid rocket engines, historically, the low burning rate of the grain or *the regression rate* has been a major challenge [8]. This difficulty has hindered the use of HREs in operations. It is known that conventional HREs, which employ polymers such as HTPB as fuel have much lower regression rates than solid rocket engines [8]. In order to tackle this challenge, liquifying fuels such as paraffin wax is utilized. These types of fuels have significantly higher regression rates due to the phenomenon called “entrainment”. Entrainment, represented in Figure 1 is a type of mass transfer mechanism, where the liquid fuel particles are transported by the oxidizer flowing over the fuel grain. The total mass transfer in conventional fuels occur primarily through vaporization. However, in liquifying fuels mass is additionally transferred through entrainment, significantly contributing to the overall mass flow rate [9].

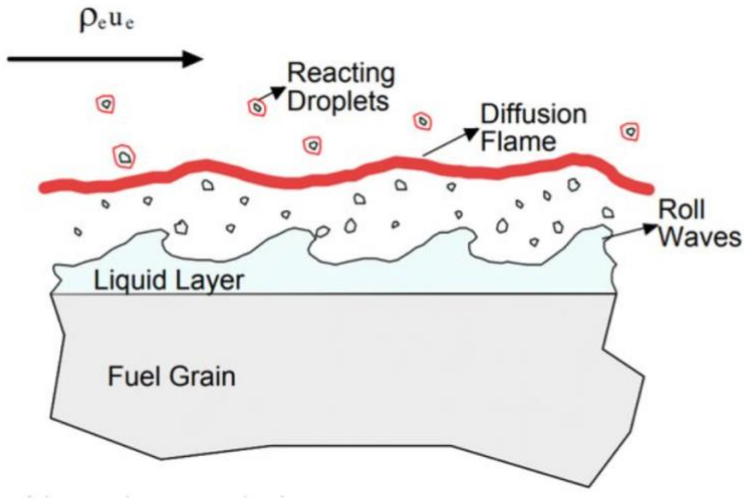


Figure 1 Mass transfer in liquifying fuels [10] [8]

Research carried out at Stanford University revealed that “The burning rate of a high-grade paraffin wax as a function of mass flux is approximately four times greater than the burning rates of HTPB. This makes the fast-burning noncryogenic wax based propellants a perfect candidate for the future generation of hybrid rockets” [8, p. 3].

The introduction of paraffin as a high-regression fuel and the consequent performance rise of HREs have contributed to renewed interest in hybrid propulsion in the last two decades. Additive manufacturing of grains has allowed for more intricate designs and improved performance. Moreover, enhanced understanding of hybrid combustion modeling, including regression rate estimation, has enabled more accurate predictions of the hybrid engine's performance. Finally, the focus on low-cost solutions for micro-launchers has resulted in renewed interest in the use of HREs. These factors combined are driving the development of hybrid rocket propulsion systems and making them an attractive option for space transportation in the future. [1]

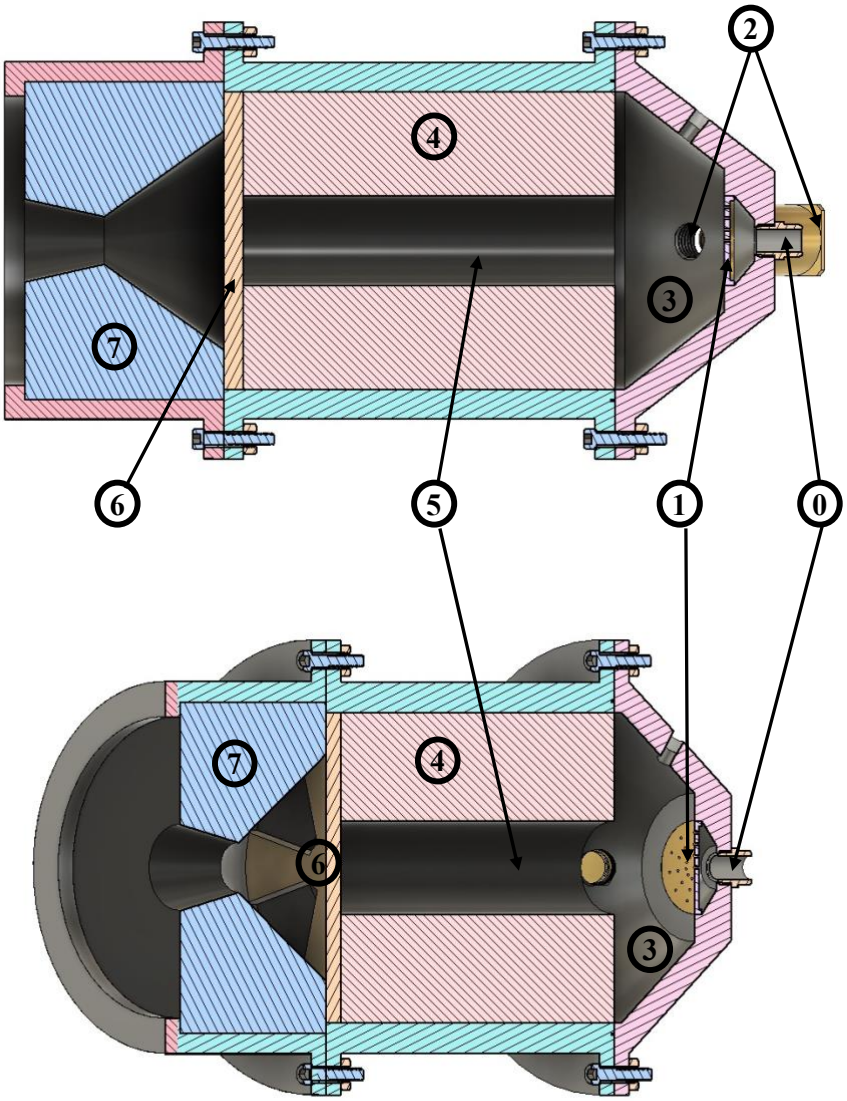


Figure 2 CAD model of a HRE (0: oxidizer inlet, 1: shower head injector plate, 2: pyrotechnical igniter/ignition port, 3: precombustion chamber, 4: single port solid fuel grain, 5: grain port, 6: postcombustion mixer, 7: conical graphite nozzle)

3.2 Hybrid Rocket Engine Design Theory

The design of HREs require a thorough understanding of various design parameters and theoretical principles. This chapter aims to provide a comprehensive overview of the fundamental concepts and mathematical formulations essential for HRE design. It begins with an exploration of key design parameters commonly employed during the development of an engine, including specific impulse, thrust, and characteristic velocity. These parameters serve as crucial metrics in assessing the performance and efficiency of HREs. The subsequent section focuses on the regression rate, a specific parameter that plays a significant role in determining

the combustion characteristics of the solid fuel in HREs. The theoretical foundations and relevant equations associated with the regression rate are discussed in detail. Finally, the chapter delves into the presentation of various fluid models employed in the design of injectors, which are vital components of HREs. These models offer insights into the flow behavior and optimization of oxidizer injection. The equations and models introduced herein are instrumental in the practical implementation of CoDeT, facilitating the efficient and accurate design of HREs.

3.2.1 Key Parameters for HRE Design

Chemical rocket engines function through the process of combusting a fuel in conjunction with a stored oxidizing agent. As presented in the Equation 3.1 [10], total mass flow of a rocket engine is determined by the sum of oxidizer and fuel mass flow rate. It directly influences the thrust and the specific impulse of an engine.

$$\dot{m}_{tot} = \dot{m}_o + \dot{m}_f \tag{3.1}$$

The *oxidizer-to-fuel (O/F) ratio* is a parameter in rocket engine design that quantifies the ratio of the oxidizer mass flow rate to the fuel mass flow rate (Eq. 3.2). It serves as the fundamental indicator of the stoichiometry and plays a crucial role in determining the heat release per unit of propellant. By specifying the O/F ratio, the chemical reaction can be controlled and the desired balance between oxidizer and fuel can be achieved, leading to high combustion efficiency and overall engine performance [11].

$$OF = \frac{\dot{m}_o}{\dot{m}_f} \tag{3.2}$$

Perhaps the most fundamental design consideration in a rocket engine development is the thrust value. Thrust can be defined as the propulsive force exerted on a vehicle by its propulsion system. In the context of rocket engines, thrust arises from the reaction force imparted on the engine structure as a result of the high-velocity expulsion of matter. This relationship is expressed by the Equation 3.3 [10] [11].

$$F = \dot{m}_{tot} v_2 + (p_2 - p_3) A_e \tag{3.3}$$

The first term in Equation 3.3, product of mass flow rate and exhaust velocity is the momentum thrust. The second term corresponds to the pressure thrust, which is calculated by multiplying the cross-sectional area at the nozzle exit by the difference between the exhaust gas pressure at the exit p_2 and the ambient pressure p_3. If the exhaust pressure p_2 is equal to the ambient pressure p_3, the rocket engine is said to have a nozzle with an optimum expansion ratio and theoretically

achieves the maximum thrust. Whereby the expansion ratio (AR) is defined in Equation 3.4 as the exit area (A_e) divided by throat area of the nozzle (A_t) [11] [12].

$$AR = \frac{A_e}{A_t} \tag{3.4}$$

In a rocket nozzle, the distribution of the actual exhaust velocity v_2 across the entire exit cross-section is non-uniform, which implies that it does not accurately represent the overall magnitude of thrust. Due to the challenge of accurately measuring the velocity profile a uniform axial velocity c is assumed. This assumed axial velocity, referred to as the *effective exhaust velocity* c, serves as the average velocity at which propellant is expelled from the nozzle. It is defined in the Equation 3.5 as the ratio of thrust to the exhaust mass flow rate [11] [12].

$$c = \frac{F}{\dot{m}_{tot}} \tag{3.5}$$

When the rocket nozzle is perfectly expanded ($p_2 = p_3$) the effective exhaust velocity c is equal to the average exhaust velocity v_2.

The *characteristic velocity*, represented as c^*, holds significant importance in rocket engine design and performance comparison. It serves as a valuable parameter that can be readily obtained through measured test data. The characteristic velocity provides insights into the combustion efficiency of the propulsion system and is notably unaffected by nozzle characteristics [13]. By considering the characteristic velocity, performance of rocket engines can effectively be analyzed and optimized. Mathematically, it can be expressed with the Equation 3.6 [12].

$$c^* = \frac{P_c A_t}{\dot{m}_{tot}} \tag{3.6}$$

By given propellant mass flow rate and characteristic velocity, chamber pressure of the engine can be calculated with the Equation 3.7 [14]. Whereby η represents the *combustion efficiency*. It indicates the ratio of the actual amount of propellant that undergoes complete combustion to the total amount of propellant supplied [14]. For a well-designed rocket engine its value ranges from 0.94 to 0.99 [11].

$$P_c = \frac{\dot{m}_{tot} c^* \eta}{A_t} \tag{3.7}$$

Another design parameter that is often used during the design process is the *thrust coefficient* c_F. Using the Equation 3.8 [10], it can easily be determined from the measured values of chamber pressure P_c, throat area A_t and thrust F. The thrust coefficient represents the degree of

thrust amplification achieved by the nozzle [15]. Higher values indicate a well-designed nozzle expansion.

$$c_F = \frac{F}{P_c A_t} \tag{3.8}$$

The definitions of characteristic velocity c^* and thrust coefficient c_F offer an alternative representation for expressing thrust [11] [10].

$$F = \dot{m}_{tot} c^* c_F = c_F P_c A_t \tag{3.9}$$

Conversely, at a given O/F ratio and chamber pressure, total mass flow rate can be calculated from known values of c^* and c_F [11].

$$\dot{m}_{tot} = \frac{F}{c^* c_F \eta} \tag{3.10}$$

The specific impulse is a fundamental design parameter used to evaluate the performance of a rocket engine. It indicates the amount of impulse produced per unit weight of propellant, thus quantifying the engine's fuel efficiency. Measured in seconds, the specific impulse is commonly employed for comparing different propulsion systems [11] [14].

$$I_{sp} = \frac{F}{\dot{m}_{tot} g_0} = \frac{c^* c_F}{g_0} \tag{3.11}$$

The initial phase of designing a rocket engine involves specifying the desired thrust level and selecting an appropriate propellant combination, along with a suitable feed system. Following this, an O/F ratio is chosen based on performance analysis and the propellant's characteristic velocity is determined. Once the characteristic velocity and mixture ratio are determined, the total propellant flow rate to achieve the desired thrust can be calculated. However, to facilitate these calculations in HREs, the introduction of a supplementary parameter is imperative. Consequently, the forthcoming section will introduce and elaborate on the concept of regression rate, which will serve as an essential factor in design process.

3.2.2 Hybrid Burning Rate Law

Combustion in the HREs is intrinsically different from the solid and liquid rocket combustion. This difference is the key in development and application of these kind of engines. Solid rockets are designed in such a way that before use, the oxidizer and fuel are blended within a single solid phase. The combustion process is initiated when the exposed surface of the solid propellant is heated to the ignition temperature by the combustion flame. In liquid rocket engines, the oxidizer and fuel are mixed in proximity to the injector to create the desired combustible mixture [11]. In contrast, HREs operate differently in that the solid fuel burns as

a turbulent diffusion flame, causing the O/F ratio to vary along the length of the combustion chamber [4]. Moreover, it is important to note that the fuel flow rate in HREs is inherently interconnected with the oxidizer flow rate and cannot be independently determined or specified [11].

The *regression rate (ṙ)* is defined as the rate at which the surface of the solid fuel grain recedes as it undergoes combustion [16]. As a fundamental design parameter, it plays a crucial role in the development and performance prediction of HREs, particularly in determining the appropriate combustion chamber length and diameter. Accurate regression rate data is crucial in comparing propellants, sizing a fuel grain, predicting the performance of an engine, and preventing burn-throughs [4].

Theories for determining the regression rate build upon the classical diffusion limited theory by Marxman in proposed in 1963 [17]. Key finding of this theory is the relationship between the local instantaneous fuel burning rate and the local instantaneous mass flux through the fuel port. In consideration of accuracy concerns with the previous theory, recently a regression rate law has been developed where the parameters of the equation are determined through sub-scale engine tests. Currently, the common way of calculating the regression rate is through an empirical power function.

The regression rate of HREs is directly linked to the oxidizer mass flux, denoted by G_o and measured in units of $\frac{kg}{s\,m^2}$. It is a readily quantifiable parameter and is defined as the oxidizer mass flow $\dot{m}_o$ divided by the average fuel port area A_p [18].

$$G_o = \frac{\dot{m}_o}{A_p} \tag{3.12}$$

The regression rate is measured $\frac{mm}{s}$. The flux coefficient *a*, along with the flux exponent *n*, are constants that are specific to the type of propellant used. Their value define dependance of regression rate on the oxidizer flux and are determined experimentally. In Table 1 values for regression rate constants are listed for different propellant combinations.

Fuel	Oxidizer	a	n
HTPB	LOX	$3.043 \cdot 10^{-2}$	0.681
HTPB-Escorez	LOX	$2.061 \cdot 10^{-2}$	0.68
HDPE	LOX	$2.340 \cdot 10^{-2}$	0.62
Paraffin	LOX	$11.70 \cdot 10^{-2}$	0.62
Paraffin	N_2O	$15.50 \cdot 10^{-2}$	0.50
Paraffin	GOX	$4.88 \cdot 10^{-2}$	0.62

Table 1: Values of regression rate coefficient *a* and *n* exponent n for various propellant combinations [19] [20]

By averaging the regression rate expression over the grain length and burn time the space-time averaged burning rate law for HREs is derived [10].

$$\dot{r} = aG_o^n \qquad 3.14$$

Using the burning rate law and the regression rate constants from Table 1, a comparison of regression rates for various fuel and oxidizer combinations can be conducted. Figure 3 illustrates the comparison of conventional non-liquifying fuels and liquifying paraffin fuel. It becomes evident that paraffin exhibits considerably higher burning rates across all propellant combinations. This phenomenon can be attributed to the entrainment process previously discussed. Due to the superior regression rate of paraffin and the subsequent high performance of HREs using paraffin-based fuels, this study will primarily focus on HREs employing paraffin as the fuel.

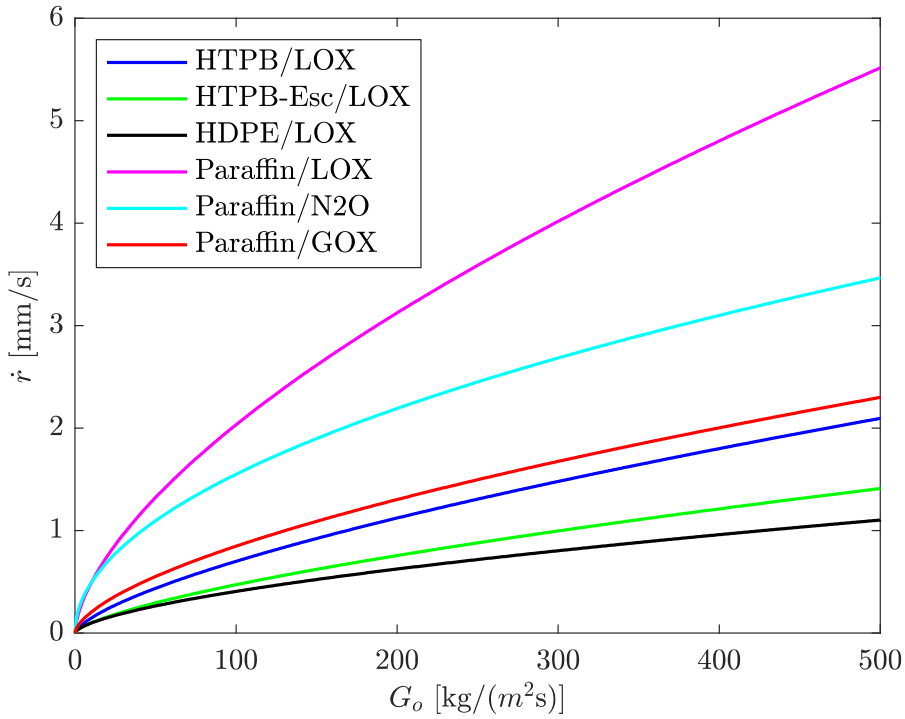


Figure 3 Regression rate of various propellant combinations at varying oxidizer mass flow rates (values for a and n are obtained from the Table 1)

The fuel mass flow rate ($\dot{m}_f$) can be determined based on the regression rate, as indicated by the Equation 3.15 [10]. It is directly related to the exposed surface area of the solid fuel grain. As the fuel port diameter gradually expands, it leads to a non-linear variation in the time-resolved mass flow rate of the solid fuel [21]. The length of the fuel grain plays a crucial role in determining the mass flow rate of the fuel, and consequently, the O/F ratio. Hence, it is essential to carefully design the geometry of the fuel grain to achieve the desired combustion characteristics. A further grain design parameter is the initial fuel port area of an engine, as it dictates oxidizer flux and regression rate throughout the combustion. During engine operation,

the fuel regresses and the fuel port undergoes expansion. This expansion leads to a reduction in the oxidizer flux, and consequently a decrease in regression rate.

$$\dot{m}_f = A_b \rho_f \dot{r} \tag{3.15}$$

3.2.3 Injector Design

The design of injectors assumes critical significance in enhancing the combustion efficiency and stability as well as engine performance of HREs [22] [23]. The selection of appropriate flow models becomes imperative in injector design to accommodate the compressibility and two-phase flow characteristics that vary based on the oxidizer type. This section introduces three such models that are utilized in mass flow rate prediction for frequently employed oxidizer types, namely GOX (gaseous oxygen), N_2O (nitrous oxide), and LOX (liquid oxygen). It draws upon the extensive work by Benjamin S. Waxman [24], whose work on oxidizer modeling and injector design of HREs serves as a fundamental reference.

In the context of HREs, the primary objectives of injectors are to introduce the oxidizer flow into the combustion chamber, atomize it into small droplets, facilitate proper propellant mixing, and thereby ensure the desired O/F ratio [11]. Various injector designs can be employed to achieve these objectives, among which the injector plate with a shower head design (SH) stands out. The SH design garners significant popularity in HREs due to its inherent advantages of simplicity in design and ease of manufacturing. It operates by dispersing the oxidizer into the combustion chamber via concentric rows of orifices which are usually drilled into the injector plate, resulting in the creation of a forceful jet entering the combustion chamber. This injector configuration is renowned for its capacity to generate high-velocity sprays. The parallel arrangement of injection ports offers advantages in terms of manufacturing simplicity. It is worth noting that the distribution and dimensions of the orifices play a crucial role in determining the spray characteristics of the SH injector [25].

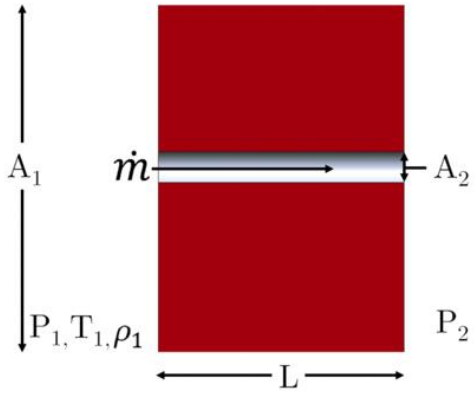


Figure 4 Injector orifice and its upstream and downstream thermodynamic conditions (Figure obtained from [24, p. 20])

In Figure 4, an SH injector plate with a single injection orifice is represented. The labels "1" and "2" are used in the following parts of the section when referencing a specific location and represent upstream and downstream conditions respectively.

3.2.3.1 Dimensionless Numbers

In order to accurately account for vena contracta (*v. c.*), which arises when a flow passes through a sharp-edged orifice and experiences friction effects, the discharge coefficient (C_d) is introduced in Equation 3.16. C_d is a dimensionless number, which quantifies the relationship between the theoretical inviscid flow without v. c. and the actual viscous flow incorporating *v. c.* [24].

$$C_d = \frac{\dot{m}_{viscid\ v.c.}}{\dot{m}_{inviscid\ no\ v.c.}} \qquad 3.16$$

The discharge coefficient of an injection orifice is influenced by factors such as its shape and manufacturing quality. Figure 5 provides reference values for different orifice types and diameters. The assessment of orifice quality necessitates conducting cold flow tests, wherein mass flow is measured using an inert liquid such as water to simulate the flow of oxidizer. To ensure accuracy, it is crucial to match the viscosity and density of the simulant fluid as closely as possible to that of the oxidizer. [11]

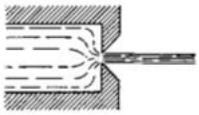
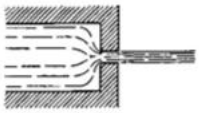
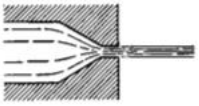
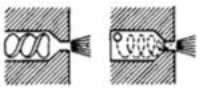
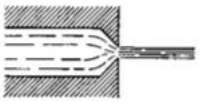
Orifice Type	Diagram	Diameter (mm)	Discharge Coefficient
Sharp-edged orifice		Above 2.5 Below 2.5	0.61 0.65 approx.
Short-tube with rounded entrance $L/D > 3.0$		1.00 1.57 1.00 (with $L/D \sim 1.0$)	0.88 0.90 0.70
Short tube with conical entrance		0.50 1.00 1.57 2.54 3.18	0.7 0.82 0.76 0.84–0.80 0.84–0.78
Short tube with spiral effect		1.0–6.4	0.2–0.55
Sharp-edged cone		1.00 1.57	0.70–0.69 0.72

Figure 5 Injector discharge coefficients for varying diameter and orifice types (Figure obtained from [11, pp. 279 Table 8-2])

Length-to-diameter-ratio (L/D) is an important characteristic of an injector orifice, which effects among others friction and *v. c.* of the flow.
The determination of L/D ratio in the design of the injector plate should consider the desired flow characteristics to minimize pressure loss. Simultaneously, the design must address operational and manufacturing limitations arising from the stress experienced by the materials utilized in the injector plate. Thus, the choice of the injector plate thickness involves a careful balance between optimizing flow characteristics and ensuring structural integrity. The equation 3.17 [24] represents the relation between L/D and C_d. C_f denotes the skin-friction coefficient, which is dependent on the Reynolds number. Additionally, K represents the pressure drop coefficient. Its value is influenced by the geometry of the system.

$$C_d = \sqrt{\frac{1}{4C_f \frac{L}{D} + K}} \qquad 3.17$$

3.2.3.2 Single Phase Incompressible Flow (SPI)

For certain oxidizers, including LOX, the influence of compressibility effects can be disregarded, and the single-phase incompressible liquid (SPI) model proves sufficient.

The SPI Model makes several assumptions: the flow is steady-state, inviscid and devoid of body forces. Furthermore, the difference in potential energy between points 1 and 2 is considered negligible, and the cross-sectional area upstream of the orifice (A_1) is significantly smaller than the orifice's cross-sectional area (A_2). Combining the continuity equation with the Bernoulli equation delivers a model without any consideration of friction losses and *v. c.*. In order to account for these effects, discharge coefficient C_d is introduced. The resulting equation is commonly referred as the C_dA*-Equation* and presented in Equation 3.18 [24].

$$\dot{m}_{SPI} = C_d A_2 \sqrt{2\rho\Delta p} \tag{3.18}$$

The density of the oxidizer is represented by ρ, while Δp denotes the pressure drop inside the orifice, which can be calculated as the difference between the upstream pressure p_1 and the orifice exit pressure p_2 . With a known oxidizer flow rate, it becomes possible to compute the total required orifice area of the injector plate ($A_{inj\,tot}$) using the Equation 3.19. Additionally, for a given number of orifices, the diameter of each orifice can be determined. [24] [11]

$$A_{inj\,tot} = \frac{\dot{m}_{SPI}}{C_d \sqrt{2\rho\Delta p}} \tag{3.19}$$

3.2.3.3 Single Phase Compressible Flow (SPC)

An oxidizer flow in gaseous state can be accurately modeled after the perfect gas. This model is based on the following assumptions. Oxidizer gas is thermally and calorically perfect, flow is isentropic and wall friction is negligible. It is also assumed that manifold flow is stationary. Under these assumptions it is possible to calculate the oxidizer mass flow rate using the Equation 3.20 [24].

$$\dot{m}_{SPC} = C_d A_2 \rho_1 \sqrt{2C_p T_1 \left[\left(\frac{p_2}{p_1}\right)^{\frac{2}{\gamma}} - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma+1}{\gamma}} \right]} \tag{3.20}$$

A_2 represents the total injector exit area and γ the ratio of specific heats. ρ_1, T_1 and p_1 are the thermodynamic conditions at injector upstream. The only relevant information required about the condition in downstream is the pressure p_2, which can be regarded as the chamber pressure of the engine.

At a certain downstream to upstream pressure ratio the mass flow rate reaches its maximum and the fluid travels with the speed of sound at the exit. The pressure ratio that corresponds to this maximum is denoted as the critical pressure ratio [14]. After this point, further increase of the pressure drop across the injector does not result in an increase in mass flow rate and the flow is said to become *choked* [24] [26]. Using the Equation 3.21 [24], critical pressure ratio can be determined from the ratio of specific heats.

$$\frac{p_2}{p_1}_{crit} = \left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma}{\gamma - 1}} \tag{3.21}$$

Under choked flow conditions, mass flow rate is solely determined by the thermodynamic conditions in the upstream region and the critical mass flow rate can be calculated with the Equation 3.22 [24].

$$\dot{m}_{SPC_{crit}} = C_d A_2 \sqrt{\gamma \rho_1 p_1 \left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{3.22}$$

3.2.3.4 Homogenous Equilibrium Model

In the case of self-pressurizing oxidizers like nitrous oxide, the pressures in the oxidizer tank and in the feed system prior to the injector are typically near saturation pressure. Consequently, as the liquid flows through an injector orifice, static pressure in the feed system drops well below the vapor pressure and significant amounts of liquid oxidizer vaporize. The flow upstream of the injector gets choked through forming of cavitating pockets [27]. The unique characteristics of two-phase flows make the previously introduced models insufficient when designing injectors and feed line systems. Here it is advised to use the *Homogeneous Equilibrium Model (HEM)* of flow.

To derive the HEM equation for mass flow rate, several assumptions are made for simplicity. Liquid and vapor travel in the system isentropic and with the same velocity. Additionally, the model assumes instantaneous heat transfer between the phases, thereby maintaining a constant state of thermodynamic equilibrium between them [28] [24]. This makes it possible to use the saturation curve when determining thermodynamic conditions.

Expression for the predicted mass flow rate using HEM ($\dot{m}_{HEM}$) is given in Equation 3.23. By the isentropic assumption ($s_1 = s_2$) the conditions downstream of the injector, density ρ_2 and enthalpy h_2 can be easily determined from the saturation curve.

$$\dot{m}_{HEM} = C_d A \rho_2 \sqrt{2(h_1 - h_2)} \tag{3.23}$$

3.3 Hybrid Rocket Engine Design Tools

This section provides an overview of the diverse array of HRE design tools found in the literature. These tools predominantly consist of software created by research teams in universities aiming to build their first hybrid propulsion system. The programming of the base software is often the first step in the development process. The code is adapted, and additions are made as needed. The data obtained during the subscale tests of the engine is used to validate the results from the model. The computational capability and user control offered by each tool varies significantly. Additionally, restricted accessibility of the source code makes it impossible to directly compare and evaluate the performance and effectiveness of different HRE design tools.

In addition to universities and research institutes, private space launch companies create and use design tools during the development of new hybrid propulsion systems. However, these tools and the associated reports are not publicly accessible, since they are protected proprietary assets. Consequently, the scope of this literature review is limited to publicly available research articles and design tools.

One notable tool is the Hybrid Rocket Performance Code (HRPC) developed by the researchers at University of KwaZulu-Natal in 2017 [29]. By B. Genevievea et al. developed tool consists of two separate components: preliminary motor design code and the predictive motor performance code. It offers users the ability to model nitrous oxide feed system, regression rate of paraffin-based fuels, and gas dynamics within combustion chamber/nozzle. Design parameters such as thrust value, chamber pressure, O/F ratio and oxidizer-fuel combination are used in the code for the design of fuel grain and nozzle configuration. Results obtained from the software are validated against the hot-fire test data and performance data of Stanford Sounding Rocket Program. HRPC is also used to design and perform analysis of a hybrid rocket demonstrator.

Sapienza University of Rome is another university, which recently (2021) developed a software intended for HRE design [30]. Researchers Paolo Maria Zolla et al. used zero-dimensional-models available in the open literature to create a code that performs fast performance analysis of HREs, helping in the rapid creation of first design iterations. It gives the user two oxidizer feed systems to choose from and compare the predicted performance of both systems. In the scope of the research work, the code is validated through rebuilding of design parameters, such as thrust value, chamber pressure and specific impulse of many publicly available static engine tests. The software demonstrated its capability by yielding satisfactory results during the validation process, affirming its reliability and accuracy.

Another recent addition to the realm of HRE design tools (2022) is the system design tool developed by the researchers in Toulouse University [2]. Their design code enables the user to simulate a specific element of the HRE system, a combination of system elements or the complete propulsion system. Different models are implemented for the simulation of feeding/injection, combustion chamber and the nozzle. The results from the program are

validated with lab-scale HRE tests and most of the results from the code are reported to match the experiment data.

It is worth noting that not only universities but also institutions like the German Aerospace Center (DLR) contribute to the development of such tools. For instance, projects *Advanced Hybrid Rocket Engine Simulation* (AHRES) and *Antriebs-Technologien für Kleinträger* (ATEK) aimed to develop a software designed to assist users in rapidly designing and optimizing hybrid propulsion systems [31]. Users can create preliminary design of an HRE tailored to the specific mission requirements or conduct predictive performance analysis of an HRE design. The software consists of many computational modules and offers the user a wide range of design choices. Among these modules is "SHAKIRA", which computes High-Test-Peroxide (HTP) decomposition. Additionally, the software includes "HEAT" module, which enables to calculate heat transfer between parts in the combustion chamber and simulate the flow of combustible gases. In the development process of a nozzle, users can use DAPHNE to obtain various parameters, including flow properties, temperature distribution and material stress. Furthermore, users have the option to incorporate regenerative cooling into their design and obtain the essential parameters required for analysis. The software further enhances the design process by offering a selection of three different grain geometries for users to choose from. In order to generate experimental data for the software, a test facility operated by DLR has been modified specifically for hybrid propulsion purposes. The facility is equipped with oxidizer supply systems for both GOX and HTP. At the time of publishing, certain software modules are undergoing validation processes through experiments conducted in this facility.

The aforementioned design tools for HREs are either closely linked to CEA (Chemical Equilibrium with Applications) or rely on input tables generated by it. Developed by NASA, CEA is a computer program specifically designed to calculate chemical equilibrium product concentrations for various reactant combinations. It offers comprehensive evaluations of thermodynamic and transport properties for the resultant mixture [32]. The program's extensive utilization in rocket engine design can be attributed to its high level of accuracy and extensive database. The design code developed in scope of this research work also relies on combustion data generated by CEA [33].

4 Methodology

4.1 NASA CEA

The Computational Design Tool (CoDeT) developed for this thesis utilizes the powerful capabilities of NASA CEA, a widely recognized program for analyzing chemical equilibrium and thermodynamic properties of rocket propellants [32]. CoDeT communicates with CEA and automates the calculations of engine design and theoretical rocket performance.

CEA can be accessed in two ways: through the *CEArun* browser-based interface or by downloading and using the *CEA GUI* [33] (Graphical User Interface). The two-part series of extensive reference publication by NASA [32] [34] , provide extensive theoretical background information on the calculations and a user manual. The explanations of CEA presented in this study are sourced from these publications.

To start with the calculations, the user is prompted to define a problem and provide the necessary specifications. For the calculation of theoretical rocket performance, the problem type is set as "rocket problem". Next, assumptions for the combustion conditions and combustor area are chosen. The user can choose to calculate performance based on chemical equilibrium and frozen composition. In the equilibrium performance, chemical equilibrium in the nozzle expansion is achieved instantaneously. Frozen performance assumes a constant product composition throughout the nozzle expansion [32]. If mass flux per chamber area or contraction ratio are not specified, default combustion chamber area is infinite (IAC). If one of the variables is specified, it is referred to as finite combustion chamber area (FAC). In the IAC, combustion is only calculated for combustion chamber diameter approaching infinite, while FAC calculates for two points, at inlet and outlet of the combustion chamber. Both IAC and FAC employ ideal-gas law. Continuity, energy, and momentum equations are reduced to one dimension. Assumptions made for the combustion chamber are as follows; propellant flow at the combustion chamber inlet is static, combustion is complete and adiabatic. Expansion of the gases in the nozzle is isentropic and gases mix homogeneously. Between the condensates and gases, temperature and velocity lags are neglected. The finite-combustion chamber of the FAC is assumed to have a constant cross-sectional area and the combustion is nonisentropic and irreversible. For IAC, the user can choose to assume equilibrium, frozen or both. For a specified combustion chamber (FAC) only equilibrium is available.

Once the combustion conditions and the combustion area defined, chamber pressure can be specified as a single value or a range with a set interval. The maximum number of datapoints is limited to 24. For propellant data, the user needs to specify fuel and oxidizer type. CEA offers a diverse catalog of propellants for solid, liquid and hybrid engines. In addition to conventional hydrocarbon fuels such as RP-1, CH_4, JP, fuel list also includes paraffin in CEA Browser version. In CEA GUI, if the desired fuel choice is not listed, the user can define a new fuel type. This is done by providing a name and specifying the enthalpy and chemical composition of the fuel. Propellant mixtures can also be created, enabling the employment of fuels with metallic additives and oxidizer mixtures. The amount of mixture elements is specified by relative mass

fraction of each element in decimals. After the propellants are selected, O/F ratio is set as a single ratio or a range of ratios with an interval. Here, the number of data points is limited to 30. For the nozzle calculations, exit conditions such as pressure ratio, sub and supersonic area ratio can be optionally specified. Like chamber pressure and O/F ratio, exit conditions can also be given multiple values. As a result, program performs calculations for every combination of specified O/F, pressure and exit condition.

Furthermore, the program allows users to specify desired output type. The available options are “short”, “long” and “debug”. Upon finalization of all user-defined input variables, conditions, and specifications, the input data is stored in a designated text file format (.inp). Subsequently, the CEA software is invoked, initiating the execution of calculations. As a result, CEA generates an output file (.out) containing the calculated values as well as the user-specified input variables. The output file serves as a repository for these stored values.

One of the significant advantages of using CEA is that it allows the user to analyze the effect of individual variables on overall rocket performance. The user can select a variable to study the effect of, such as the O/F ratio, and define an interval they would like to analyze, while keeping other variables constant. CEA calculates the performance parameters for each O/F ratio and through this, allowing for a quantitative assessment of its impact on performance.

Although the results obtained from the program are accurate and extensive, extracting the relevant information from the large output file can be challenging. Another limitation of CEA, both in browser and GUI forms, is the limited number of chamber pressure and O/F ratio data points (24 and 30, respectively). A more detailed analysis requires a higher number of data points, highlighting the need for automated program execution and extraction of desired information.

To address these challenges, CoDeT utilizes the CEAwin function, which acts as a bridge between CoDeT and CEA. The MATLAB function CEAwin, authored by Cyrus Foster [35], automates the execution of CEA. Specifically, the function writes the input file of CEA for the user input, executes CEA, retrieves the results, and extracts data from the output file. Data is stored in a structure variable named “rocket_perf”. This structure variable is used to organize and store the various output values obtained from the CEA program. The output data is assigned to specific fields within the rocket_perf structure, making it easier to access and retrieve the desired information.

In the following section algorithm of CoDeT will be explained in detail.

4.2 Algorithm

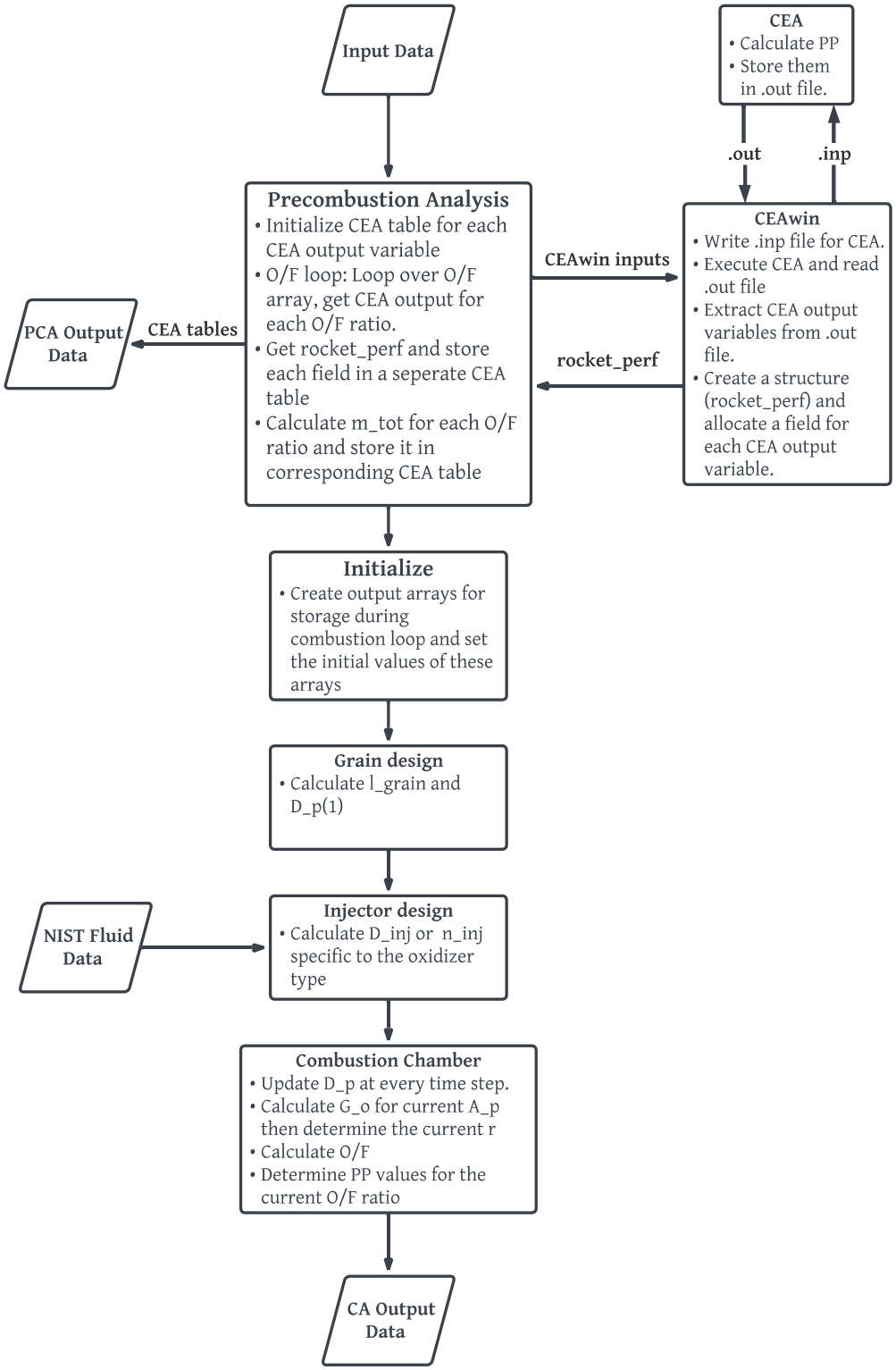


Figure 6 Flow chart of CoDeT. (PP: Performance Parameters, PCA: Precombustion Analysis, CA: Combustion Analysis, variable names are listed in Table 2 and 3)

CoDeT algorithm can be divided in three main parts:

1) Precombustion analysis ($t \leq 0$): Analysis of performance parameters and O/F, initialization and choosing the initial O/F ratio, grain design, injector design
2) Combustion modeling ($0 < t \leq t_{burn}$): calculate performance parameters and regression rate during combustion
3) Postcombustion analysis ($t > t_{burn}$): store combustion data and plot results

4.3 Input Parameters

Input parameters of CoDeT consist of CEAwin inputs, used for calling the function and inputs, which are used for calculations inside CoDeT. In this section both types of input parameters will be explained.

CEAwin function provides CEA with all the necessary information for the calculations. Some input variables and conditions required by CEA are set as constants in function algorithm and can only be changed by modifying the code. Others, variables listed in Table 2 are function inputs for CEAwin and are assigned by the user.

CEAwin input variable	Unit/Type
Name fuel additive (Addt)	str
Name primary oxidizer (Ox_1)	str
Name secondary oxidizer (Ox_2)	str
Relative weight paraffin (m_par)	-
Relative weight primary ox. (m_ox1)	-
Relative weight additive (m_addt)	-
Chamber pressure (P_c)	bar
Area ratio (AR)	-
Temperature paraffin (T_par)	Kelvin
Temperature primary ox. (T_par)	Kelvin
Temperature additive (T_add)	Kelvin
Ambient pressure (p_amb)	psi

Table 2 CEAwin input variable names and input units

CEAwin is called by defining the reactant types, combustion conditions and exit conditions. User can introduce reactants by creating fuel and oxidizer blends consisting of two reactants. Fuel consists of base fuel type, which is already set as paraffin in CEAwin. Paraffin is listed in reactant catalog of CEA and the software stores the reactant enthalpy of -1860.6

$kJ/(kg\ mol)$ at 298,15 *K* and chemical formula $C_{32}H_{66}$. Here it is important to mention that the provided reactant enthalpy is valid for temperature at and around 298,15 *K*. User can choose to employ the following performance enhancing additives: BH_3NH_{33}, $LiAlH_4$, AlB_{12}, MgB_2. Oxidizer blends are created with two elements which are called the primary and secondary oxidizers. Composition of the propellant elements are set by relative weight. O/F ratio is defined in the function as the relative weight of fuel divided by the relative weight of the oxidizer. For instance, if the fuel choice is pure paraffin and the used oxidizer only consists of one type, by

setting relative weights of paraffin and primary oxidizer to one and two respectively, an O/F ratio of two is achieved. With the information on chamber pressure, combustion condition of the reactants is fully defined. The exit conditions, that need to be assigned for the function is the area ratio (AR) and the ambient pressure (p_amb). CEA calculates two types of specific impulse. Vacuum specific impulse, which is calculated for neglected ambient pressure and specific impulse (I_{sp}), which is calculated using the assigned ambient pressure.

In addition to input variables used to execute CEA, in Table 3 listed variables are assigned by the user for calculations inside CoDeT. Thrust output, combustion efficiency, oxidizer flux and the fuel density are used in combustion chamber modeling. Hereby the calculations are iterated over a burn time array. The information on *a* and *n* define the regression rate behavior of the propellant combination and must correspond to the chosen reactant types (see Tab. 1 for exact values). For the propellant combination, the code iterates over the user specified range of O/F values. This allows the analysis of performance parameters in relation to O/F. The exact operations are explained in the following section.

CoDeT input variable	Unit/Type
Desired thrust output (F_des)	N
Combustion efficiency (comb_eff)	-
Initial oxidizer flux (G_o_initial)	$kg/(s\ m^2)$
Grain outer diameter (D_p_max)	mm
Burn time (t_burn)	s
Time increment (dt)	s
Injector orifice diameter (D_inj)	mm
O/F start value (OF_st)	-
O/F end value (OF_end)	-
O/F increment (OF_inc)	-
a	-
n	-
Fuel density (rho_f)	kg/m^3

Table 3 CoDeT input variables and respective units

4.4 Precombustion Analysis

Precombustion encompasses the initial operations conducted in CoDeT before the combustion modeling phase. This section initiates with an examination of performance parameters, specifically focusing on their variations with changing O/F ratios. Subsequently, the code proceeds with initialization, followed by the design of the grain and the injector.

For the examination of performance parameters, CoDeT implements an iterative process (O/F loop) over a user-defined array of O/F ratios, invoking the CEAwin function in each iteration. Within this loop, the CEA input file is dynamically modified for every O/F value, and the CEA program is executed accordingly. The resulting performance data, encapsulated within the structure variable rocket_perf, is extracted and stored in separate tables corresponding to the output parameters of interest.

Upon completion of the O/F loop, the code accumulates the performance data associated with each O/F ratio. Specifically, the tables contain values for important metrics such as specific impulse (I_{sp}), characteristic velocity (c*), thrust coefficient (C_F), and chamber temperature, with each value correlated to its corresponding O/F ratio. These tables are referred to as *CEA tables*.

Furthermore, utilizing the obtained performance parameters, the code calculates the total mass flow for each O/F iteration using the equation 3.10. This comprehensive dataset can subsequently be subjected to analysis, visualization, and further calculations or comparisons to gain insights into the engine's behavior under various O/F ratios.

4.4.1 Initialization

The initialization section of the code performs the necessary setup and initialization steps before starting the iterative calculations in the combustion chamber modeling.

Firstly, arrays are initialized for the burn time array to store data related to propellant mass flow rates, oxidizer flux, grain geometry, and other quantities. Calculations in the combustor are performed at every time step and stored in the initialized arrays presented in Table 4. Burn time array starts at zero seconds and stops on the user defined burn time. The burn time array is initialized starting from zero seconds and continues until the defined burn time is reached. The time step size remains constant throughout.

Combustion Array	Stored data
m_tot	Total mass flow rate
m_f	Fuel mass flow rate
m_o	Oxidizer mass flow rate
r	Regression rate
G_o	Oxidizer flux
F	Thrust
A_p	Port area
A_b	Burn area
D_p	Port diameter
OF	Oxidizer-to-fuel-ratio
CF_perfect	Thrust coefficient
Isp_perfect	Specific impulse
cstar	Characteristic velocity
T_chamber	Chamber temperature

Table 4: Combustion arrays of CoDeT and respective data

The code then identifies the O/F ratio associated with the maximum specific impulse from the CEA table for I_{sp}. This ratio is used as the initial O/F ratio for the combustion. Initial values for the variables such as total mass flow, thrust coefficient, specific impulse, characteristic velocity, and chamber temperature values corresponding to the initial O/F ratio are selected from the CEA tables.

Furthermore, the initial values for fuel mass flow, oxidizer mass flow are calculated from the initial total mass flow rate and O/F ratio (refer to Eq. 3.2). The initial flow rate of the oxidizer does not change throughout the burn time.

Using the equation 3.14, initial regression rate is calculated from regression rate constants and initial oxidizer flux.

All output arrays are created and initial values for the parameters at the start of burn time ($t = 0$) are set.

4.4.2 Grain Design

The geometry of the paraffin fuel modeled by CoDeT is a single port cylindrical grain. Outer diameter of the grain (D_p_max) is specified by the user. Initial port diameter (D_p) and grain length are calculated by CoDeT.

Together with user provided initial oxidizer flux (G_o), initial port diameter of the fuel grain (D_p) is calculated.

The code proceeds to calculate the grain length (l_grain) required to sustain the initial O/F ratio. It iteratively determines the grain length by adjusting it until the calculated O/F ratio matches the desired initial O/F ratio within a given tolerance. In each iteration, the burn area is calculated based on the initial port diameter and current grain length. The fuel mass flow rate is determined using the equation 3.15. The O/F ratio is computed by dividing the oxidizer mass flow rate by the current fuel mass flow rate. The loop continues until the O/F ratio is within the predefined tolerance. The grain length is then stored as the variable l_grain and is given as output.

4.4.3 Injector Design

Injector Design Input Parameters	Unit/Type
Used model for the design	SPI/SPC/HEM
Calculated output type	n_inj_hole/D_inj
Pressure drop (deltaP)	bar
Discharge coefficient (c_d)	-
Heat capacity ratio (gamma)*	-

Table 5 Injector design input parameters with respective units (*only required for SPC model)

The injector plate is designed specific to the chamber pressure, the oxidizer type inputs, and the oxidizer mass flow rate determined during initialization phase. The region upstream of the injector (orifice inlet) is referred to as the *manifold*, and the pressure in this area is determined by the given pressure drop. So, across the injector, deltaP amount of pressure drops from manifold to combustion chamber pressure.

For each model, thermodynamic conditions of the oxidizers are sourced from NIST Chemistry WebBook fluid data [36].

The code section for injector design begins by prompting the user to choose one of the three models incorporated within CoDeT: single phase incompressible (SPI), single phase compressible (SPC), or Homogenous Equilibrium Model (HEM). In each model after the calculation of the total injector area based on equations specific to the respective model, user is prompted to determine output type. If the type "n_inj_hole" is selected, user is asked to determine the orifice diameter. Conversely, in the case of output type "D_inj", number of orifices needs to be set by the user.

If the user selects injector design based on SPI, the code proceeds to calculate the required injector hole diameter or the number of injector holes with the equation 3.18. Density of the liquid oxidizer is obtained from the chamber pressure and vapor quality, which is assumed to be zero.

In the injector design based on SPC, in contrast to previous model a conditional statement is utilized, which determines if the oxidizer flow is chocked and calculates the total injector orifice area accordingly. The condition for the choked flow is set by the equation 3.21. If the condition is true, calculations are performed based on the equation 3.22. Oxidizer density under upstream conditions is determined from the manifold pressure and steam mass fraction, which is assumed to be one. In the case of unchoked flow, the equation 3.20 is utilized. The required thermodynamic conditions are once again determined from the pressure and vapor quality in the manifold.

The algorithm for the HEM approach starts with determining the thermodynamic conditions in inlet (manifold) and outlet of the orifice then uses these values in the equation 3.23 for calculation of output parameter. Upon completing this section of the code, the user is provided with the minimum required number of orifices or the diameter of these orifices, which are values, specific to the chamber pressure and oxidizer mass flow rate.

4.5 Combustion Analysis

The section of the code pertaining to combustion chamber modeling involves an iterative process that simulates the combustion process over a series of time steps. Within this iteration, various calculations are performed to determine output parameters.

The first calculation as represented in the Equation 4.1 involves updating the port diameter (D_p) based on the regression rate (r) from the previous time step. This allows for tracking the expansion of the port as combustion progresses. The port diameter (D_p) at the current time step (n) is determined using the following equation:

$$D_p^n = D_p^{n-1} + 2 \cdot \dot{r}^{n-1} \cdot dt \qquad 4.1$$

Since the regression rate is typically expressed in units of millimeters per second, multiplying it by the time step size (dt) ensures that the change in port diameter aligns with the time progression.

After calculating the port diameter, a check is conducted to verify if the fuel has been completely depleted. This is done by comparing the current port diameter to the input grain outer diameter (D_p_max). If the port diameter exceeds this threshold, indicating fuel depletion, the error "*Fuel has run out at t_end seconds. Increase D_p_max or G_o or decrease t_burn*" is generated to notify the user. With "t_end" indicating the exact time of fuel depletion.

The port area (A_p) and burn area (A_b) are then computed using the current port diameter and the grain length.

Current oxidizer mass flux is calculated by dividing the oxidizer mass flow rate with the current port area. Its value decreases over time due to the increased cross-sectional area available for the constant rate of oxidizer flow. Regression rate is calculated with the burning rate law (Eq. 3.14).

The fuel mass flow rate (m_f) is calculated by considering the burn area, fuel density, and current regression rate (refer to equation 3.15). Furthermore, the total mass flow rate (m_tot) is obtained by adding the fuel mass flow rate to the oxidizer mass flow rate. The current O/F is determined by dividing the oxidizer mass flow rate by the current fuel mass flow rate.

Using the current O/F ratio in the loop, corresponding values of performance parameters are interpolated from CEA tables.

Finally, the thrust (F) is computed using the interpolated values of C_F, total mass flow rate, characteristic velocity, and the input combustion efficiency (refer to Eq. 3.9). This calculation yields an estimation of the thrust generated by the combustion chamber at the current time step.

By repetitively executing these calculations for each time step, the code effectively models the behavior of the combustion chamber throughout the specified time interval.

4.6 Postcombustion Analysis

The resulting combustion arrays are stored in a table file and the values are plotted against time. The tabulated data can be further analyzed, visualized, or employed for subsequent calculations or comparative studies.

5 Results and Discussion

5.1 Introduction

In this section, the capabilities of CoDeT are demonstrated through a comprehensive study of theoretical performance and combustion modeling. The investigation focuses on the impact of input parameters on engine performance and design, utilizing CoDeT to analyze and predict the resulting data. The findings and insights gained from this analysis are thoroughly discussed, providing valuable information on the behavior and characteristics of HREs under various operating conditions.

In the precombustion analysis, theoretical performance parameters and combustion variables are computed for three engines utilizing diverse oxidizers in combination with paraffin fuel ($C_{32}H_{66}$). CoDeT data is employed to investigate the variations in performance parameters under different mixture ratios, leading to a comprehensive exploration of trade-offs associated with each oxidizer choice. Subsequently, the impact of a crucial nozzle design parameter, namely the area ratio, on the overall engine performance is examined, and a design choice is made based on the obtained results.

For each engine, an injector design is conducted, taking into account the initial conditions derived from the precombustion analysis. Then, in the combustion analysis, the theoretical performance of the engines throughout the burn time is simulated using CoDeT data. This modeling highlights the operational differences and performance variations between different oxidizers, contributing to a deeper understanding of the engines' behavior during combustion.

5.2 Precombustion Analysis

5.2.1 Part 1: Theoretical Performance Comparison of Propellants

The analysis data aims to shed light on the advantages, limitations, and trade-offs associated with each oxidizer choice. For this reason, all design parameters (as specified in Tab. 6) except the employed oxidizer are kept consistent among the compared engines.

Thrust output and chamber pressure are deliberately selected to represent a HRE that is within the realm of possibility for construction in research settings. The chosen values are in line with operational requirements that can be met by small research groups and universities, making the analysis more practical and applicable to real-world scenarios.

Fuel	Ox_1	m_par [-]	m_addt [-]	P_c [bar]	F [N]	AR [-]	T_par [Kelvin]	T_ox [Kelvin]	p_amb [atm]	Comb_eff- [-]
$C_{32}H_{66}$	GOX	1	0	30	1000	1	298.15	293	1	0.98
$C_{32}H_{66}$	LOX	1	0	30	1000	1	298.15	90	1	0.98
$C_{32}H_{66}$	N_2O	1	0	30	1000	1	298.15	293	1	0.98

Table 6 Paraffin/GOX, LOX and N_2O input parameters

The comparison involves three HREs utilizing the most common oxidizer types for hybrids: liquid oxygen (LOX), gaseous oxygen (GOX), and liquid nitrous oxide (N_2O). These engines are all powered by pure paraffin with the chemical formula $C_{32}H_{66}$ (Dotrioacontane) and a density of 1120 kg/m^3. Throughout this study, these engines will be referred to by their respective oxidizer choices.

For each engine, a comprehensive theoretical performance analysis is conducted, exploring various mixture (O/F) ratios with a step size of 0.1. To cover a wide range of potential mixtures, an O/F range of 0 to 10 is selected. This range is well-suited for the specific propellant combinations, as it includes the commonly employed mixture ratios for these HRE configurations.

The area ratio (AR) is initially set as one, and its impact on performance will be assessed in subsequent examination. The combustion efficiency is predetermined to represent a well-designed engine. The analysis is conducted under a sea-level pressure condition of 1 atm. Frozen equilibrium assumptions are employed for all calculations conducted.

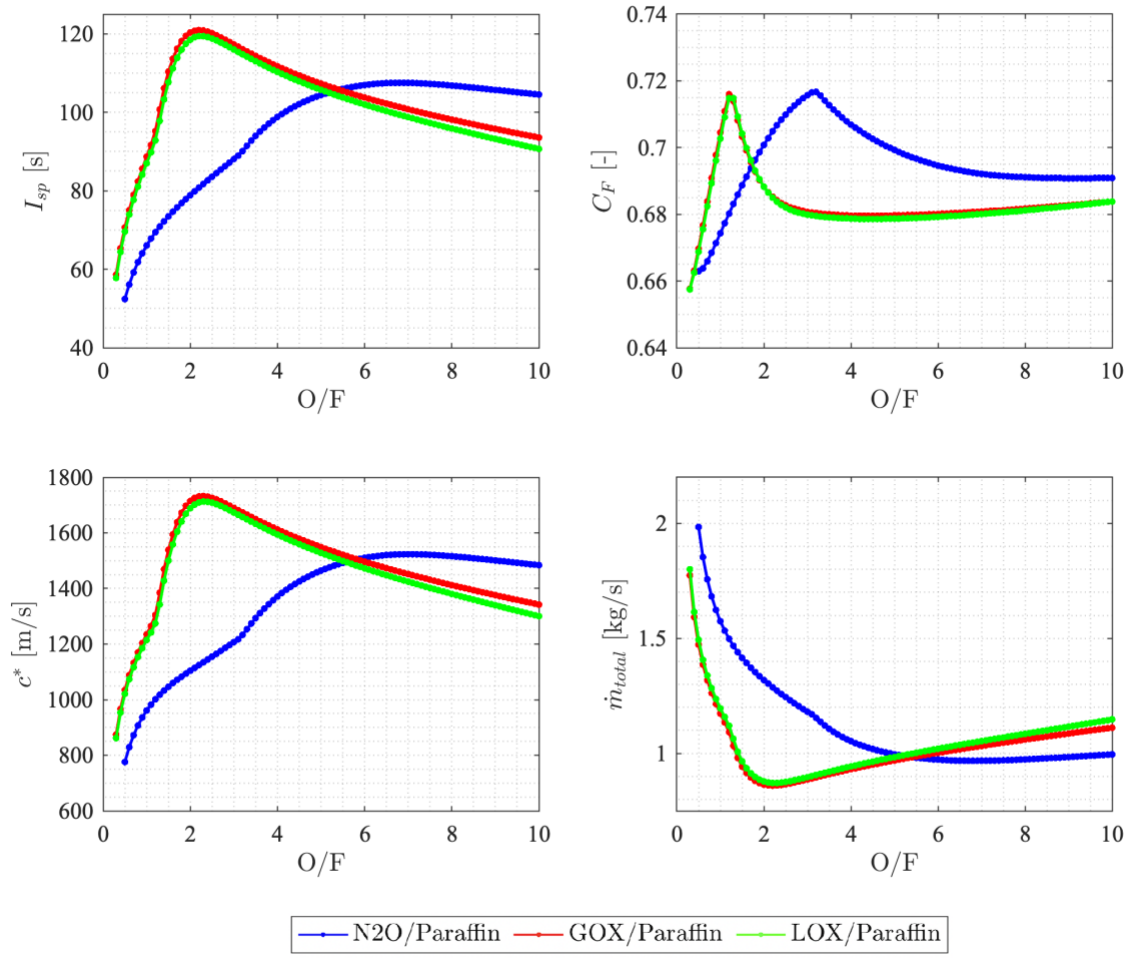


Figure 7: Performance parameters of three propellant combinations under varying O/F ratios

(Chamber pressure of 30 bar, *AR=1*, total mass flow rate calculated for a 1kN engine)

The output data of precombustion analysis for N_2O, GOX and LOX engines are presented in Figure 7. Upon examination of the data presented, it becomes apparent that the performance of both the LOX and GOX engines show minimal variations between them. In contrast, the performance of the N_2O engine stands out distinctly, exhibiting significant disparities when compared to LOX and GOX engines.

The liquid and gaseous oxygen engines achieve the highest specific impulse performance, approximately 120s. In comparison, the N_2O engine attains a maximum I_{sp} that is 11% lower than oxygen engines. It is noteworthy that the maximum specific impulse values are attained at significantly different O/F ratios, specifically 2.3 for oxygen and 6.8 for N_2O.

At their respective maximum I_{sp} points, oxygen hybrids require approximately 12% less total mass flow rate compared to N_2O hybrids. Since all three engines deliver the same thrust, this indicates that oxygen hybrids yield higher specific thrust (thrust per unit mass flow rate) than N_2O hybrids.

Given that the exit conditions are uniformly set for each engine, the maximum thrust coefficient values are nearly identical among them. The investigation of the thrust coefficient's dependence on the area ratio will be conducted in subsequent analysis.

The performance parameter often chosen when comparing propellant performance is the characteristic velocity, since it is only slightly depended on ambient pressure, chamber pressure and area ratio. Achieved maximum c* of oxygen hybrid is 12% greater than N_2O hybrid. Maximum characteristic velocity values for oxygen and N_2O correspond almost exactly to the O/F ratios with highest I_{sp} performance. For ease of reference, these O/F ratios (2.3 for GOX and LOX and 6.8 for N_2O at an area ratio of 1) will be referred to as the "optimal O/F ratios" from this point forward, as they are the mixtures at which the engines deliver the highest I_{sp} and near maximum c* performance.

Based on the obtained results of maximum characteristic velocity and specific impulse performance, it is evident that the highest overall performance can be achieved by employing a HRE with GOX or LOX as the oxidizer. However, apart from performance considerations, the selection of oxidizer type is also influenced by operational requirements specific to each oxidizer. Therefore, a comprehensive evaluation of both performance and operational factors is crucial in selecting the most suitable oxidizer for a given propulsion system.

The choice of oxidizer is a delicate trade-off between maximizing performance and ensuring system simplicity. Different missions may have unique demands, and the budgetary constraints also play a significant role in decision-making. As a result, the propulsion system can be optimized either for superior performance or for a simpler and cost-effective design, depending on the specific use-case.

The operational characteristics of the three compared oxidizers exhibit notable distinctions.

LOX is stored at cryogenic temperatures of 90K. This quality presents challenges and significant costs in terms of its management and storage during engine testing. Due to its extremely low temperatures, specialized systems are required to maintain the oxidizer at these conditions throughout the testing process.

To deliver the required oxidizer mass flow rates, LOX requires the implementation of either a turbopump or a pressurizing gas tank. Both systems add complexity and weight to the overall propulsion system. However, these additions may be justified for larger systems with high thrust output and weight, as the added weight of the oxidizer feed system becomes relatively low compared to the total weight of the propulsion system.

In contrast, GOX and liquid N_2O can be stored at room temperature and operate in blow-down mode, eliminating the need for external pressurization. This property makes their use preferential for small to medium scale propulsion systems.

Notably, N_2O has a unique self-pressurizing property. However, this property can only be exploited at a narrow temperature/pressure window. Consequently, under certain ambient conditions, nitrous oxide tanks may require external heating. Additionally, it is essential to

address the potential hazard associated with the decomposition of nitrous oxide, which requires the implementation of specific safety precautions to ensure safe and reliable operation.

	GOX (293 K)	LOX (90 K)	N_2O (293 K)
Density at 50 bar [kg/m^3]	68.6	1152.6	785.1

Table 7 Density of oxidizers [28]

The density of the oxidizing agents plays a crucial role in determining the weight and size of the propulsion system. In Table 7, the densities of each oxidizer at a tank pressure of 50 bar are provided. From these values, it becomes evident that there is a significant density difference between gaseous and liquid oxidizers. Liquid oxidizers use the propulsion system volume more efficiently, as they require smaller tanks for the same amount of oxidizer.

To achieve performance levels similar to LOX (as shown in Fig. 7), the GOX system necessitates large tanks. The size of the tanks can be reduced to some extent by utilizing a high tank pressure, this in turn necessitates thicker tank walls, increasing the system weight.

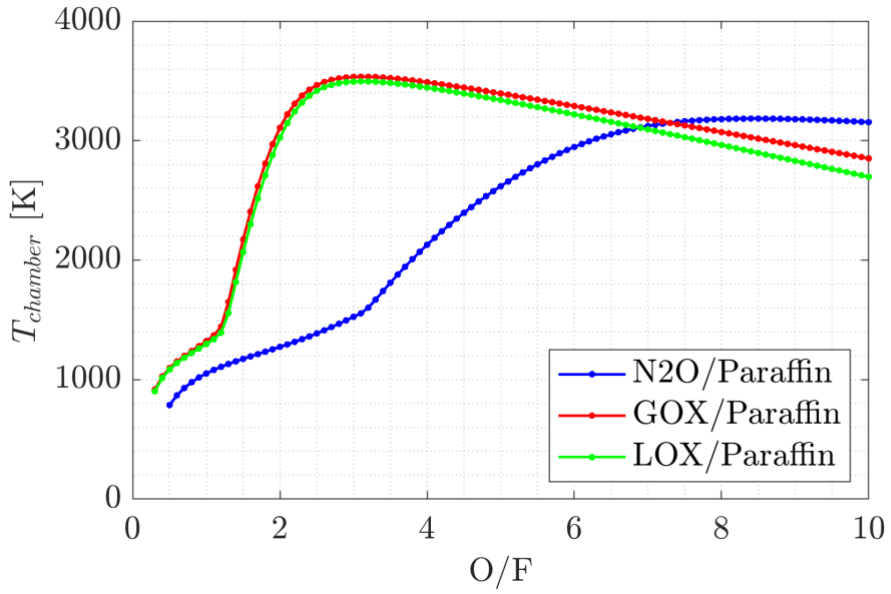


Figure 8 Adiabatic Flame Temperature in the combustion chamber (Chamber pressure of 30 bar)

Another parameter that will be investigated is the adiabatic flame temperature.

The adiabatic flame temperature serves as a crucial parameter that offers valuable insights into an engine's combustion characteristics. As depicted in Figure 8, the maximum flame temperature of GOX and LOX is achieved at an O/F ratio of 3.1, reaching approximately 3500 K. In contrast, N_2O exhibits a significantly lower maximum flame temperature of 3185 K at an O/F ratio of 8.5. At their respective optimal O/F ratios, the temperatures are not much lower than the maximum values, specifically 3300 K for GOX and LOX, and 3100 K for N_2O. This observation underscores the trade-off wherein achieving high performance comes with the consequence of elevated chamber temperatures.

The combustion chamber and nozzle materials are subjected to extreme stresses during combustion, particularly at temperatures exceeding 3000K. To achieve reusability of engine components and utilize cost-effective materials, it is advantageous for the engine to operate at a lower flame temperature while still maintaining satisfactory performance. This approach ensures that the materials can withstand the harsh conditions and prolong the lifespan of the engine components, reducing overall costs and improving efficiency.

The stoichiometric ratio, which represents the mixture ratio at which complete combustion takes place [14], is calculated to be around 3.4 for the reaction of paraffin ($C_{32}H_{66}$) with oxygen and approximately 9.5 with N_2O. This reference value is crucial in determining whether the engine operates in an oxidizer-rich (fuel-lean) or fuel-rich mode. The choice between these two modes depends on various design factors, and either can be deemed desirable depending on the specific requirements of the system.

One concern that arises is the oxidation of materials used in the combustion chamber and nozzle. To mitigate this issue, designers can opt for a fuel-rich mixture ratio. Another aspect to consider in HREs is the erosion of the nozzle during engine operation. Due to the high concentration of oxidizers, chemical erosion can occur in HRE nozzles, leading to alterations in the engine's performance characteristics. Employing a fuel-rich mixture ratio, can help decrease the erosion rate at the nozzle throat due to the reduced presence of highly reactive oxidizing agents [37].

In the case of HREs, the O/F ratio is subject to change during engine operation and cannot be controlled as precisely as in the case of liquid rocket engines. As a result, the combustion within the engine can shift modes, transitioning to a higher O/F ratio, which corresponds to an oxidizer-rich condition. This phenomenon and its implications will be thoroughly discussed in the combustion analysis

5.2.2 Part 2: Effect of Area Ratio on the Theoretical Performance

Fuel	Ox_1	m_addt [-]	P_c [bar]	AR [-]	T_par [Kelvin]	T_ox [Kelvin]	p_amb [atm]	Comb_eff [-]
$C_{32}H_{66}$	GOX	0	30	1-5	298.15	293	1	0.98
$C_{32}H_{66}$	LOX	0	30	1-5	298.15	90	1	0.98
$C_{32}H_{66}$	N_2O	0	30	1-5	298.15	293	1	0.98

Table 8 Input Parameters of the three engines for studying the effect of area ratio on performance

To find the optimal area ratio specific to the given chamber pressure and ambient pressure, performance parameters are calculated for increasing ratios. The results of these calculations are then compared. This performance analysis considers a range of area ratios that are typically used in engineering models (EM) for testing to characterize the engine's performance.

Area ratio of one serves as the base line for this analysis and represents an engine without any expansion of combustion gases. Due to the exit condition specified as an ambient pressure of 1

atm, the area ratio values being compared are relatively low. A HRE used in higher-stage rockets or orbital applications would necessitate a significantly higher area ratio for achieving optimal expansion of the combustion gases. While this performance study does not specifically investigate the performance of nozzle expansion ratios under various ambient pressures, it is important to consider the engine performance across a range of ambient pressures that the engine operates at.

As previously mentioned, characteristic velocity remains constant regardless of the area ratio. Likewise, adiabatic flame temperature does not change with increasing AR, therefore, its value is not further examined. The focus is directed towards other relevant parameters that exhibit variation with the AR changes to gain insights into the engine's efficiency under different nozzle exit conditions.

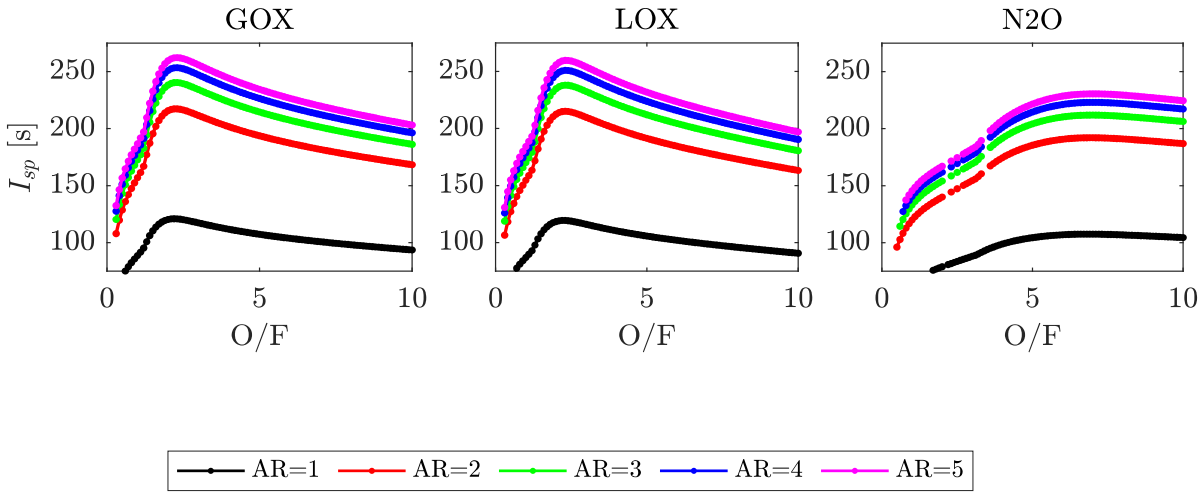


Figure 9 Change of specific impulse with AR (at 30 bar chamber pressure and 1 atm ambient pressure)

In Figure 9 several common characteristics between oxidizers are observed. With same thrust output, an increase in AR delivers a more efficient engine with higher specific impulse. This improvement in efficiency can be attributed to a more complete expansion of the combustion gases. Increase in performance diminishes with each incremental increase in the area ratio. The optimal O/F ratios for each oxidizer wander to slightly higher O/F ratios, while the required total mass flow rate decreases. With an increase of AR from 2 to 5, it is possible to reduce the required total mass flow rate by approximately 17%. This improvement in propellant efficiency leads to an overall lighter propulsion system, thereby increasing the payload capacity.

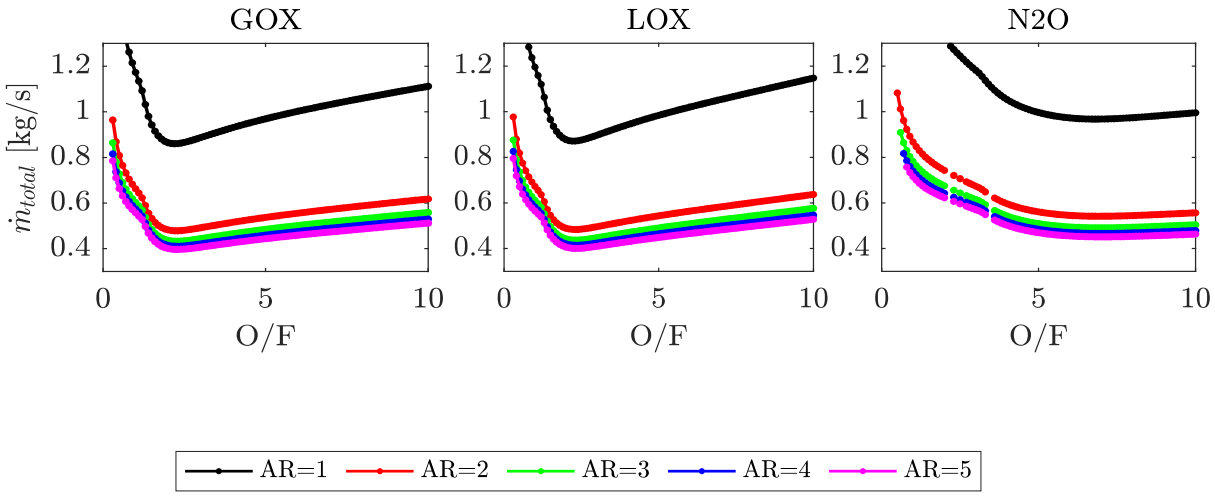


Figure 10 Change of total mass flow rate with AR (at 30 bar chamber pressure and 1 atm ambient pressure)

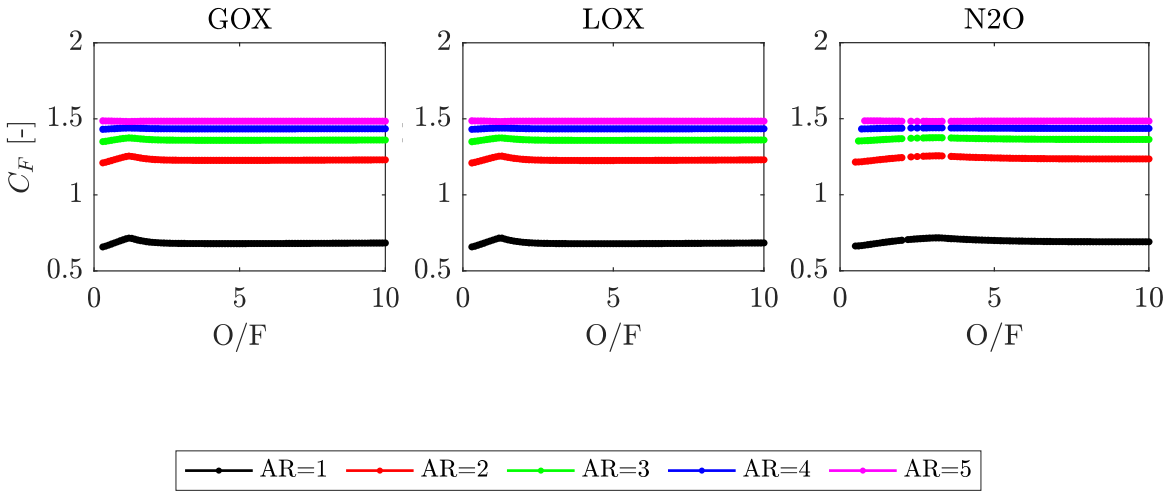


Figure 11 Change of thrust coefficient with AR (at 30 bar chamber pressure and 1 atm ambient pressure)

The thrust coefficient serves as a valuable parameter for assessing the nozzle's design efficiency. Its value represents how much in the combustion chamber produced thrust is amplified by the nozzle [15].

The thrust coefficient values show consistency across different oxidizer types. As the area ratio increases, the amplification of thrust becomes more pronounced. Specifically, transitioning from an AR of 1 to 2 results in a significant increase in the thrust coefficient, indicating that the nozzle begins to amplify the thrust generated in the combustor. This trend continues as the AR increases further, and at an area ratio of 5, the nozzle enhances the thrust by 50% compared to the thrust produced in the combustor alone.

The thrust coefficient value of 1.5 signifies a well-designed rocket nozzle [15] that has been optimized for the specified operating conditions. CoDeT's calculations demonstrate that this specific value is achieved at an area ratio of 5 for all oxidizer types. Consequently, this area ratio is chosen as the nozzle specification for the combustion analysis. By employing this particular area ratio, the engine is expected to perform with high efficiency and thrust amplification during combustion.

5.2.3 Injector Design

	GOX	**LOX**	**N_2O**
Oxidizer mass flow rate [kg/s]	0.276	0.279	0.395
Used flow model	SPC	SPI	HEM
Pressure drop [bar]	15	15	20
Number of injector orifices	5	5	5

Table 9 Injector design input parameters

For the calculated oxidizer mass flow rates of GOX, LOX and N_2O engines (presented in Tab. 9), the required injector orifice diameters for each engine are determined for a SH injector plate with 5 and 10 orifices. The results are presented in Figures 12 and 13, where their variation is plotted against discharge coefficient of the injector orifices.

The pressure drop values at the injector are standard values commonly used in HREs when the chamber pressure is set at 30 bar. Notably, the pressure drop for the nitrous oxide injector is higher, specifically at 20 bar. This intentional design choice is made to ensure that nitrous oxide remains in a liquid state in the oxidizer tank, which typically requires a pressure level of approximately 50 bar at 298K.

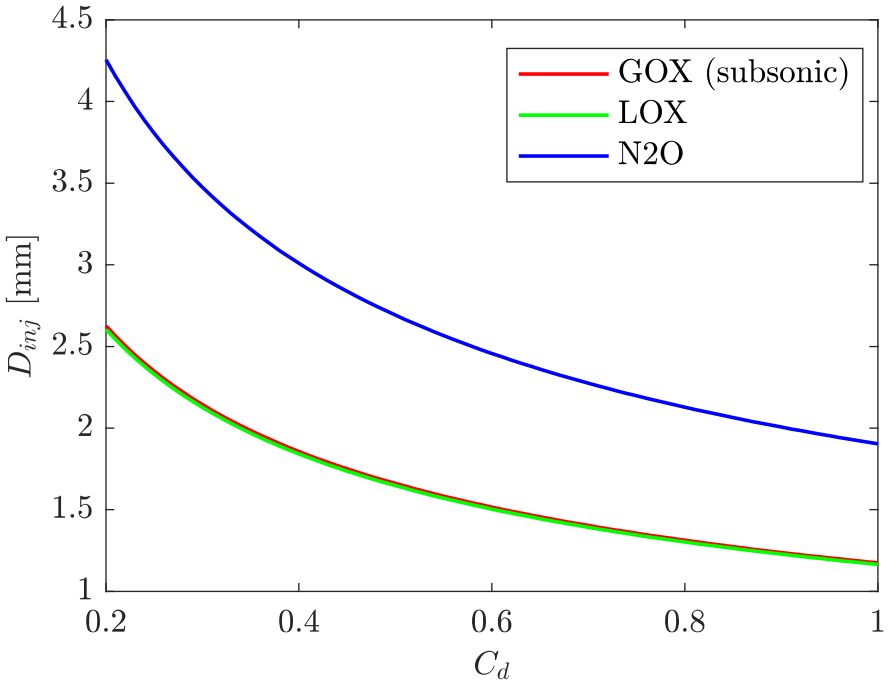


Figure 12 Injector orifice diameter of three oxidizer flows at various discharge coefficient values (SH injector plate with 5 orifices)

Figures 12 and 13 show that the required orifice diameters for GOX and LOX flow are very similar. In contrast, the diameter for nitrous oxide flow is notably higher due to the higher mass flow rate associated with this specific engine configuration.

At the provided injector pressure drop of 15 bar for GOX, the flow is subsonic, which means the oxidizer flow is not choked and the mass flow rate is dependent on the chamber pressure.

At the specified GOX injector conditions, choked flow is encountered when the upstream pressure of 58 bar is reached. However, at this pressure oxygen flow becomes supercritical. As a consequence, the calculations of CoDeT for the injector design become infeasible.

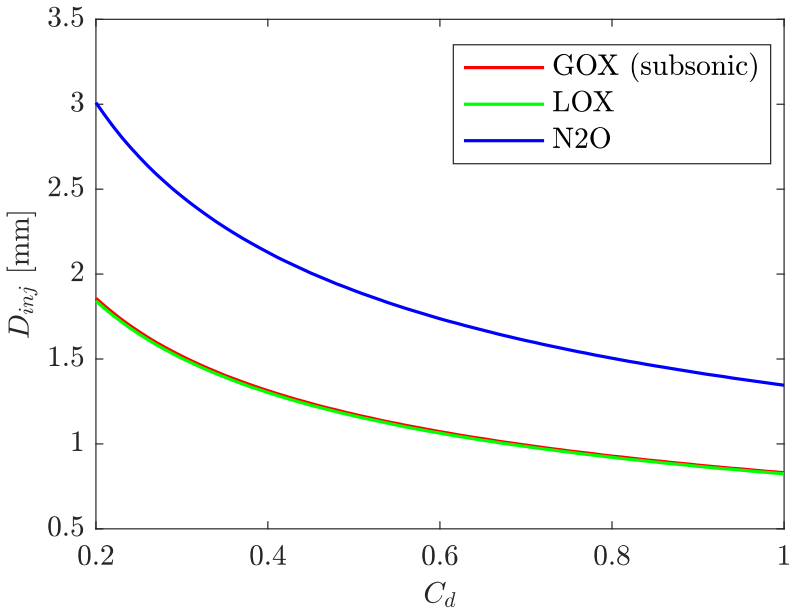


Figure 13 Injector orifice diameter of three oxidizer flows at various discharge coefficient values (SH injector plate with 10 orifices)

For ease of manufacturing, standardized bore diameters such as 1, 1.5, and 2mm are chosen. Because the nitrous oxide flow requires larger diameters for the same discharge coefficient, 1.5 and 2 mm are chosen. For each oxidizer bore diameter pairs, a targeted discharge coefficient value is determined with CoDeT results. The outcome of this approach is presented in Table 10.

	GOX/LOX		**N_2O**	
Number of orifices	5	10	5	10
Discharge coefficient	0.6	0.68	0.9	0.8
Diameter [mm]	1.5	1	2	1.5

Table 10 Targeted discharge coefficient values at preferential orifice diameters

Based on discharge coefficient and orifice diameter values provided by CoDeT, theoretical suggestions for the design of injector orifices can be made. For a hybrid engine utilizing a SH injector plate with 10 orifices, it is recommended to use a sharp-edged cone type orifice for the oxygen flow (refer to figure 5 for recommended orifice types for various discharge coefficients). For the nitrous oxide flow, a short tube with a conical entrance type orifice can be employed.

However, it is crucial to emphasize that these suggestions are based on theoretical models with a series of assumptions. In practice, real values of the discharge coefficient and mass flow rate may differ significantly from the theoretical predictions. Therefore, it is crucial to perform cold flow tests to accurately determine these values and optimize the design of the injectors for efficient and reliable performance of the HRE.

5.3 Combustion Analysis

In this section, the engine specifications in Table 8 are supplemented with a set of combustion parameters, which are detailed in Table 11.

The selected initial oxidizer fluxes for each engine are within a range of values commonly utilized by engines with similar design parameters. In the combustion loop each value of output arrays are calculated at a frequency of 100Hz. Values for *a* and *n* are set for the specific propellant combination.

The engines start their operation with their respective optimal O/F ratios (2.3 for GOX/LOX and 7 for N_2O). Initial values of performance parameters (I_{sp}, c*, C_F, $\dot{m}_{tot}$) are set as the values corresponding to these ratios. Moreover, it is assumed that the designed injector plates in the injector design section provide a constant oxidizer mass flow rate throughout the entire burn time.

CoDeT input variable	**GOX**	**LOX**	**N_2O**
Thrust [N]	1000	1000	1000
Combustion efficiency	0.98	0.98	0.98
Chamber pressure [bar]	30	30	30
Ambient pressure [atm]	1	1	1
Area ratio	5	5	5
Initial oxidizer flux [$kg/(s\ m^2)$]	300	300	200
Burn time [s]	10	10	10
Time increment [s]	0.01	0.01	0.01
a	0.049	0.12	0.16
n	0.62	0.62	0.5

Table 11 Input parameters for combustion modeling

In Table 12 calculated design parameters of the fuel grain and the results of the combustion modeling are presented.

Oxygen hybrids require a higher fuel mass to deliver nearly the same thrust output. This is because they operate at a much lower O/F ratio than N_2O hybrid. While nearly the same fuel mass is required for GOX and LOX hybrids, fuel grain of the GOX engine is notably longer. This difference can be attributed to the considerably higher regression rate of LOX in comparison to GOX, which results in the need for a significantly larger surface area to maintain the same initial O/F ratio for the GOX engine (refer to Fig. 1).

As in the precombustion analysis predicted, mean performance of the oxygen hybrids are higher. Their values during the combustion will be discussed in the following.

Grain design	**GOX**	**LOX**	**N_2O**
Initial port diameter [mm]	34.2	34.4	50
Final port diameter [mm]	57.5	77.9	83.1
Grain length [mm]	583	245	145
Used fuel mass [g]	1095	1056	560
Burn time averaged performance parameters			
Thrust [N]	968	952	999
Oxidizer flux [kg/(s*m2)]	169.16	115.95	115.65
O/F	2.53	2.66	7.04
Oxidizer mass flow rate [kg/(s)]	0.276	0.279	0.395
Fuel mass flow rate [kg/(s)]	0.109	0.105	0.056
Regression rate [mm/s]	1.16	2.17	1.65
I_{sp} [s]	261	257	230
Thrust coefficient	1.485	1.485	1.485
c* [m/s]	1724	1701	1523
Adiabatic flame temperature [K]	3468	3471	3127

Table 12 Output parameters combustion analysis for a 1 kN engine operating at sea-level ambient pressure with a chamber pressure of 30 bar AR=5

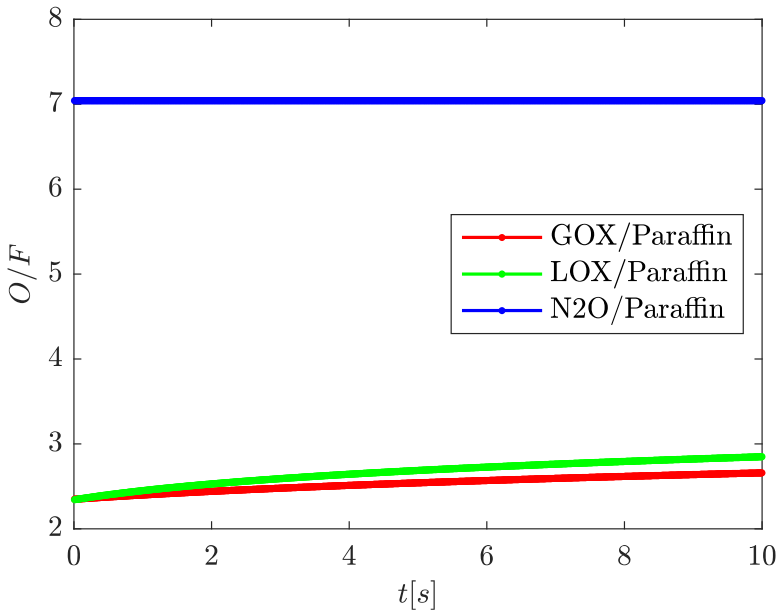


Figure 14 O/F Shift of engines

The combustion analysis results demonstrate a notable difference in the combustion behavior between oxygen and nitrous oxide when paired with paraffin. Specifically, the mixture ratio of oxygen hybrids undergoes changes over time during the burn, while the N_2O engine maintains a constant initial ratio throughout the combustion process.

This phenomenon of changing O/F ratio in HREs commonly referred to as the "O/F shift", is typically observed in single-port HREs with a constant mass flow rate. The exception are the engines with a mass flux exponent (n) of 0.5 [38], which is the case for the N_2O engine.

According to results from CoDeT (as visualized in Figure 14) in a burn time of 10 seconds, O/F ratio increases from the optimal O/F ratio by 0.3 and 0.5 for GOX and LOX engines, respectively.

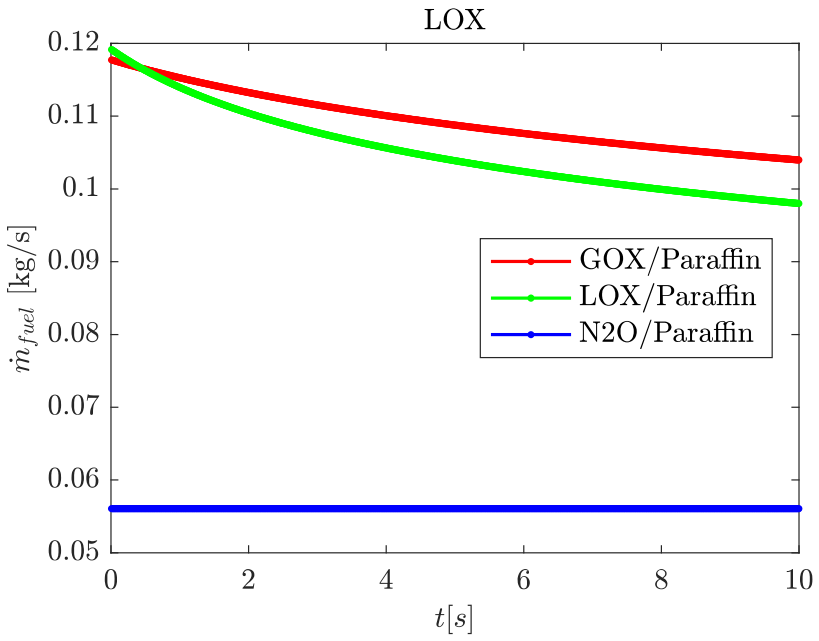


Figure 15 Change of fuel mass flow rate with time

To gain a deeper understanding of this observation, the changes in the fuel mass flow rate and regression rate among the three hybrid engines need to be examined. Since the oxidizer mass flow is constant, the increase in O/F ratio in oxygen hybrids can be attributed to the decrease in the fuel mass flow rate. As expressed in Equation 3.15, rate of fuel expelled during combustion is a function of burn area and regression rate.

As depicted in Figure 16, the regression rates of all hybrids decrease as the port expands, which should result in a reduction of fuel mass flow rate. However, in the case of the N_2O engine, the decrease in regression rate is perfectly offset by the increase in burn area. On the other hand, the increase in burn area for oxygen hybrids does not keep pace with the reduction in regression rate, leading to a decrease in the fuel mass flow rate and subsequent increase in the O/F ratio.

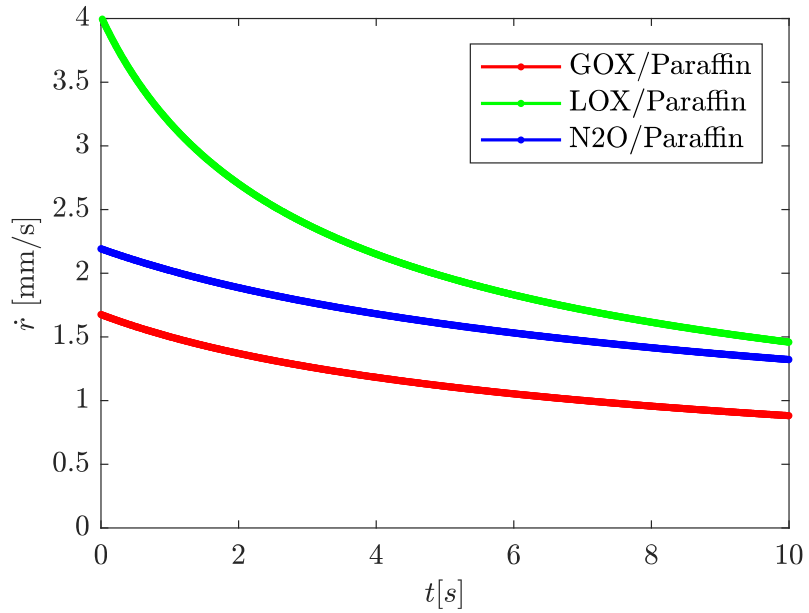


Figure 16 Temporal change of regression rate of three engines

This distinction in the O/F ratio behavior between oxygen and N_2O engines carries implications for the combustion process and the overall performance characteristics of the HREs. As the O/F ratio deviates from the initial optimal value, the engine's performance output becomes less than its maximum potential. This can be seen on the temporal change of thrust, specific impulse and c* presented in Figures 17, 18 and 19.

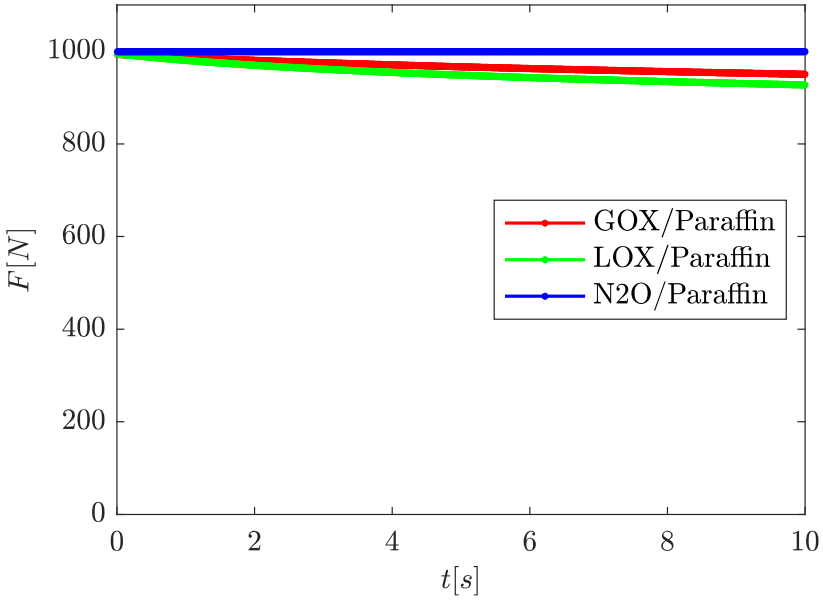


Figure 17 Temporal thrust change of three engines

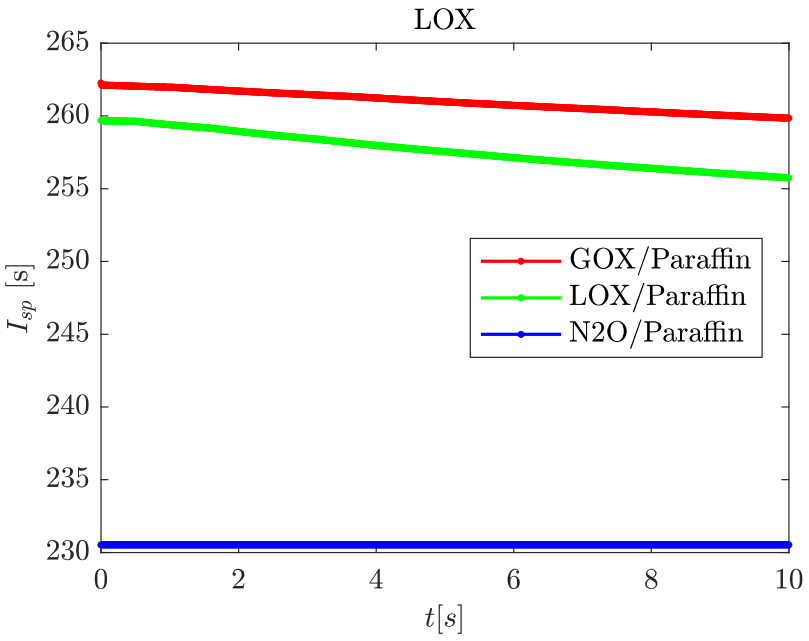


Figure 18 Temporal change of specific impulse for three engines

The drop in characteristic velocity and specific impulse during the combustion phase for GOX is less than 0.1%. Similarly, for LOX, the decrease in c* and I_{sp} is around 1.5%. These reductions in performance values are so minute that they are almost impossible to resolve with the accuracy of the commonly used mass flow measurement methods during engine test [38]. As a result, the deviations in performance parameters due to variations in O/F ratio can be considered minor, and the overall performance of oxygen hybrids proves to be stable and consistently higher than that of N_2O engines.

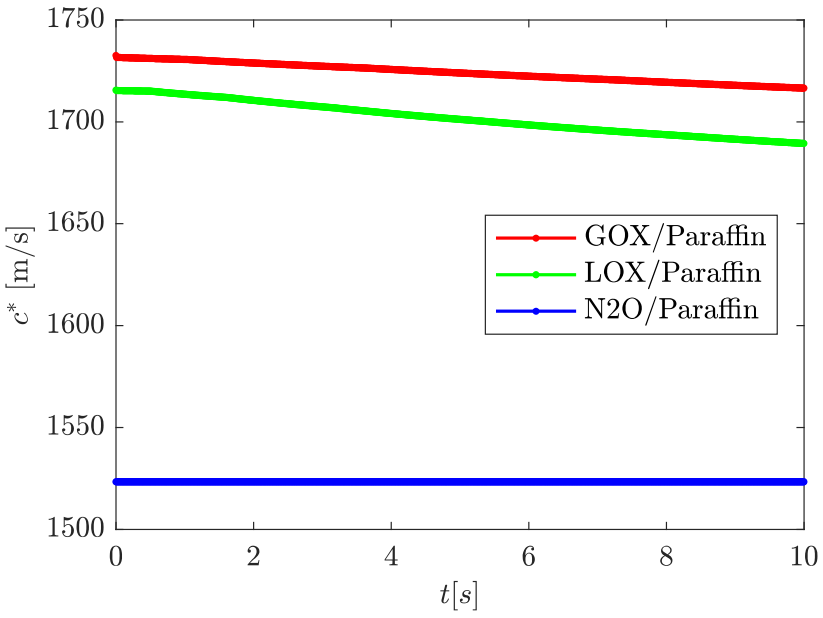


Figure 19 Temporal change of characteristic velocity

5.4 Validation of the Results

5.4.1 Theoretical Performance Validation with RPA:

The CoDeT results presented in the section *5.2.2 Effect of Area Ratio on the Theoretical Performance* (Figures 8, 9, 10, and 11) are subjected to validation using *Rocket Propulsion Analysis (RPA)* v.1.2.9 Lite Edition. RPA is a commercially available computational package intended primarily for use in the preliminary design of liquid propellant rocket engines [3]. However, the software's fuel catalog also includes solid fuels, including paraffin wax, allowing its applicability for HREs.

The combustion performance analysis is conducted in RPA based on the engine specifications of three engines as provided in Table 8. The three engines are defined within the RPA software to closely match the specifications used in CoDeT to ensure comparability of results.

In the following, engine definition process in RPA software is detailed.

The specification of chamber pressure for the analysis is standardized at 30 bar. The propellant specification involves a bipropellant configuration, comprising three oxidizers and a fuel, all selected from the reactant catalog. The oxidizers include gaseous oxygen at 298.15 K, liquid oxygen at 90 K, and nitrous oxide at 298.15 K. Additionally, the selected fuel is $C_{32}H_{66}$ (Dotriacontane), a solid propellant, with a heat of formation value (-967.8 kJ/mol). These values in RPA are sourced from NIST Chemistry WebBook.

In the nozzle flow model, the analysis considers the expansion area ratio, ranging from 2 to 5. However, it should be noted that the calculations for an area ratio of 1 have encountered issues, leading to the exclusion of performance validation at this particular ratio from the overall validation process. The flow condition is assumed to be in frozen equilibrium, and the reaction efficiency is estimated based on the predefined engine parameters. Moreover, the nozzle is configured as a bell nozzle, and its efficiency is estimated using the specified nozzle exit condition. Ambient pressure is set at 1 atm.

Performance parameter values of I_{sp} and thrust coefficient obtained from the RPA software are the operational values. Because total mass flow rate is not specifically calculated by RPA, its value is determined outside the software from the performance parameters with a combustion efficiency of 0.98, which is the same value given to CoDeT.

In the following, RPA and CoDeT output data are compared. The relative discrepancy values are given in percent and are relative to the CoDeT values.

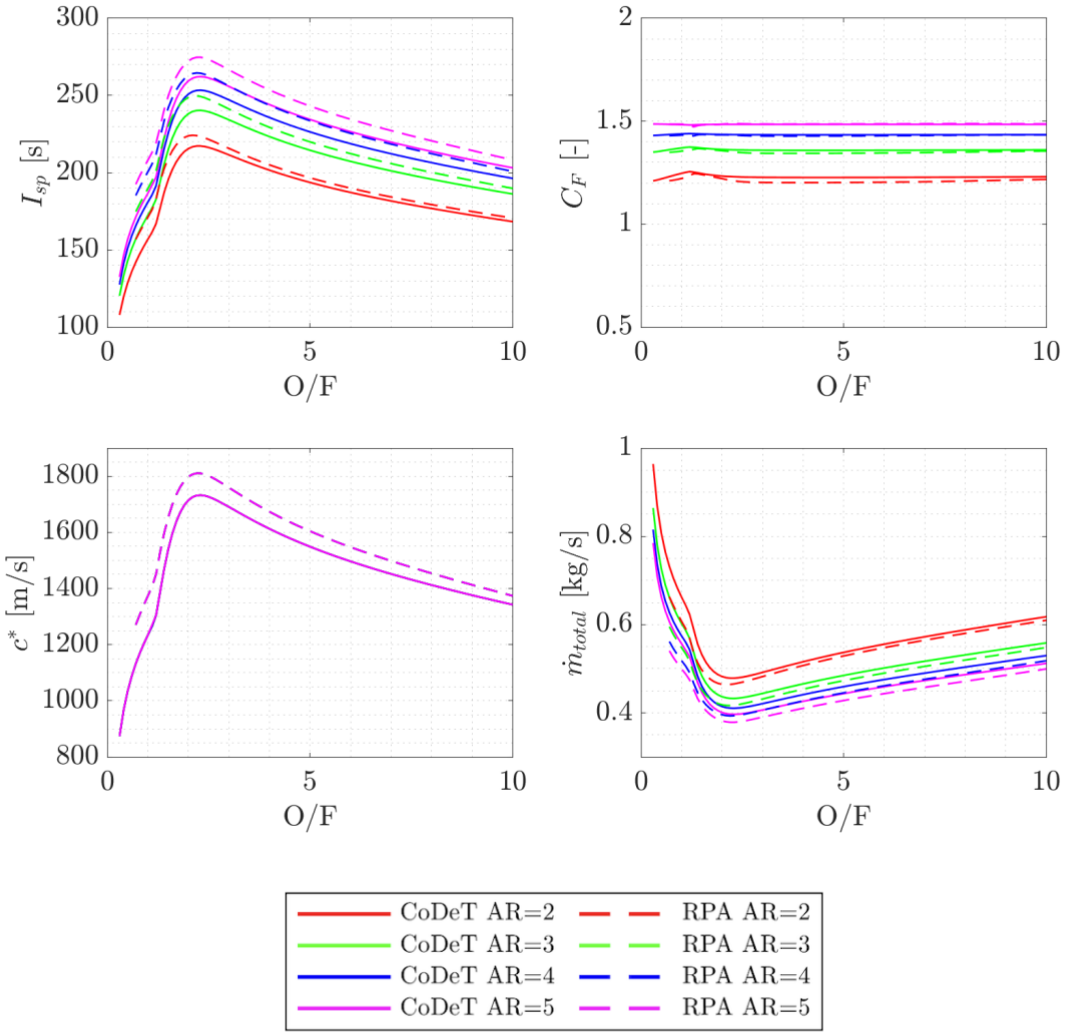


Figure 20 Validation of precombustion analysis data with RPA GOX (Chamber pressure of 30 bar with a thrust output of 1kN)

The comparison between CoDeT and RPA results for GOX reveal specific observations related to the I_{sp} and characteristic velocity values at different area ratios and O/F ratios.

As the AR increases, the mean discrepancy of I_{sp} values between CoDeT and RPA also increases, ranging from 2.5% to 4.3%. This indicates that the two methods diverge in their predictions as the engine operates at higher AR values.

The discrepancies between RPA and CoDeT results also exhibit variations with increasing O/F ratios. For GOX and LOX data, as the O/F ratio increases, the differences in predictions become more similar between the two programs, with an approximate 3% discrepancy observed at O/F ratios starting from 2.5. Notably, around the O/F ratio mark of 2, a distinct decline in discrepancy values is observed.

In contrast, the thrust coefficient values show a significant overlap and exhibit good agreement between CoDeT and RPA, with practically no discrepancy.

Furthermore, similar to I_{sp} the characteristic velocity calculated by RPA consistently exhibits higher values compared to CoDeT, with a mean discrepancy of 4.1%.

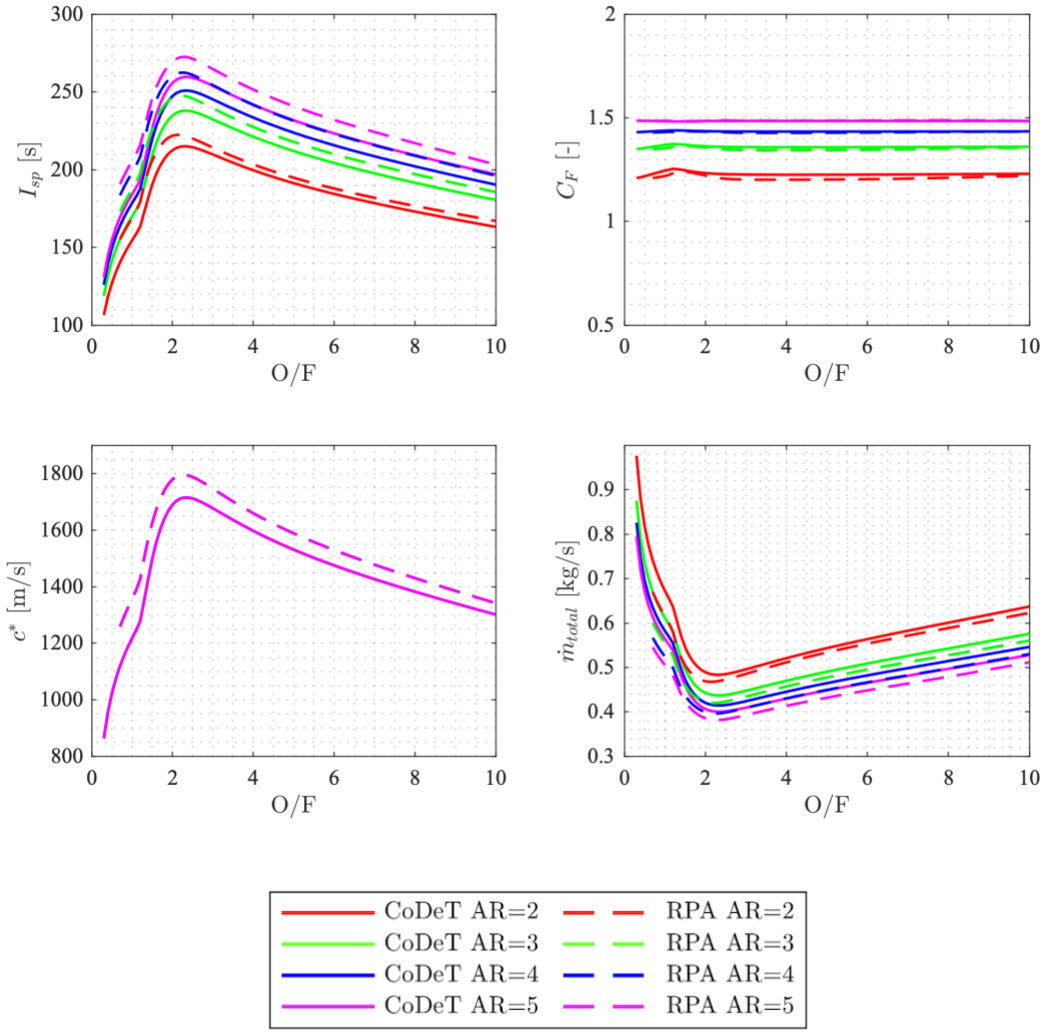


Figure 21 Validation of precombustion analysis data with RPA LOX (Chamber pressure of 30 bar with a thrust output of 1kN)

Similar to CoDeT, RPA delivers marginally lower performance values for LOX in comparison to GOX.

The discrepancy of LOX data between CoDeT and RPA is slightly higher with a mean discrepancy of 3 to 4.7%.

The values of optimal O/F ratio indicate an agreement between CoDeT and RPA data as they predict the same ratio of 2.3 for GOX and LOX. However, as previously mentioned there are differences in the calculated I_{sp} and c* values at this optimal O/F ratio between CoDeT and RPA. Specifically, for GOX at this ratio, CoDeT predicts an I_{sp} of 262s, while RPA predicts 275s, resulting in a 5% discrepancy. Similarly, the characteristic velocity c* is 4.5% higher in RPA (1810) than in CoDeT (1732).

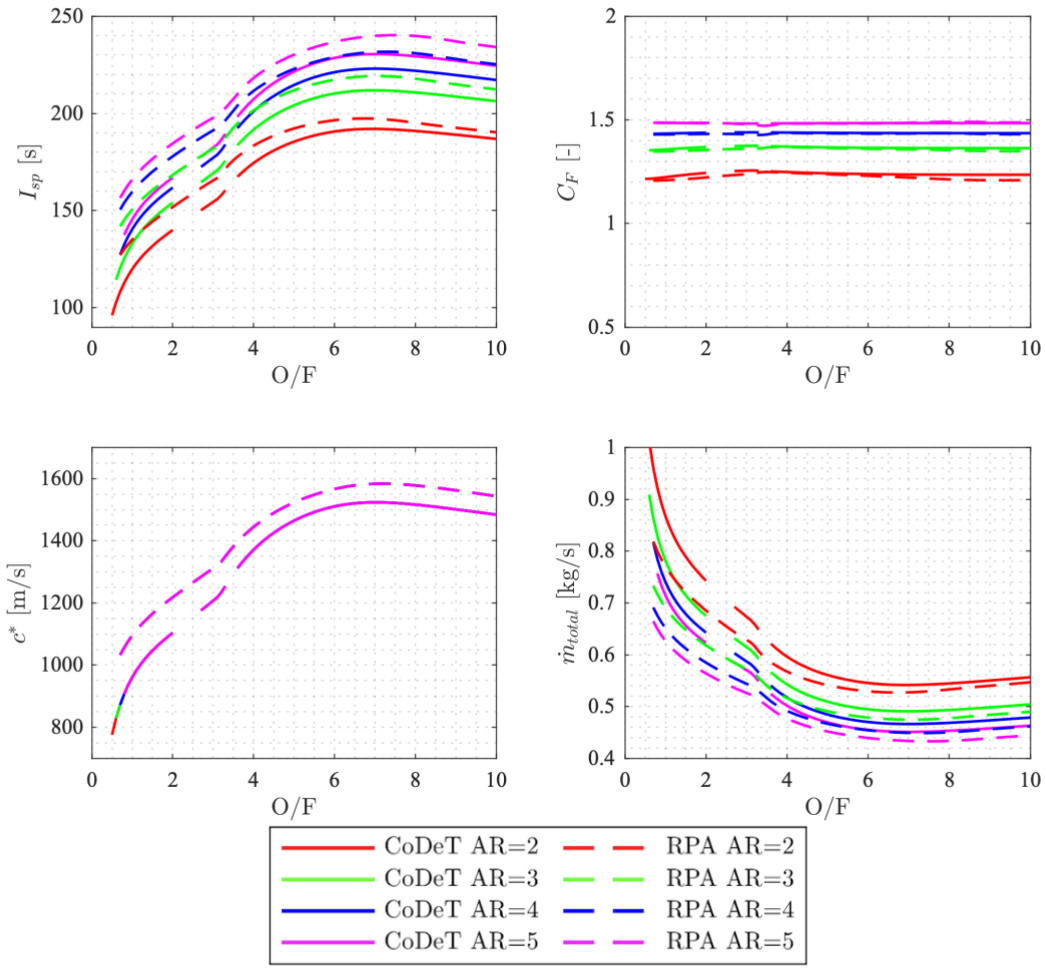


Figure 22 Validation of precombustion analysis data with RPA N_2O (Chamber pressure of 30 bar with a thrust output of 1kN)

The mean discrepancy of I_{sp} for N_2O is higher compared to GOX and LOX, with a range of 4.7% to 5.9%. Similar to GOX and LOX, the I_{sp} discrepancy decreases with increasing O/F ratio, indicating a trend of convergence between CoDeT and RPA predictions at O/F ratios higher than 2. However, unlike GOX and LOX, this decrease in I_{sp} discrepancy for N_2O is more gradual.

Notably, the optimal O/F ratio for N_2O differs significantly between RPA and CoDeT. RPA calculates an optimal O/F ratio of 7.5, while CoDeT yields a value of 7. At the optimal O/F ratio, RPA predicts an I_{sp} of 240s, while CoDeT predicts a lower I_{sp} of 230s.

Moreover, the mean discrepancy of c* for N_2O is higher, with a 6% mean discrepancy between CoDeT and RPA results.

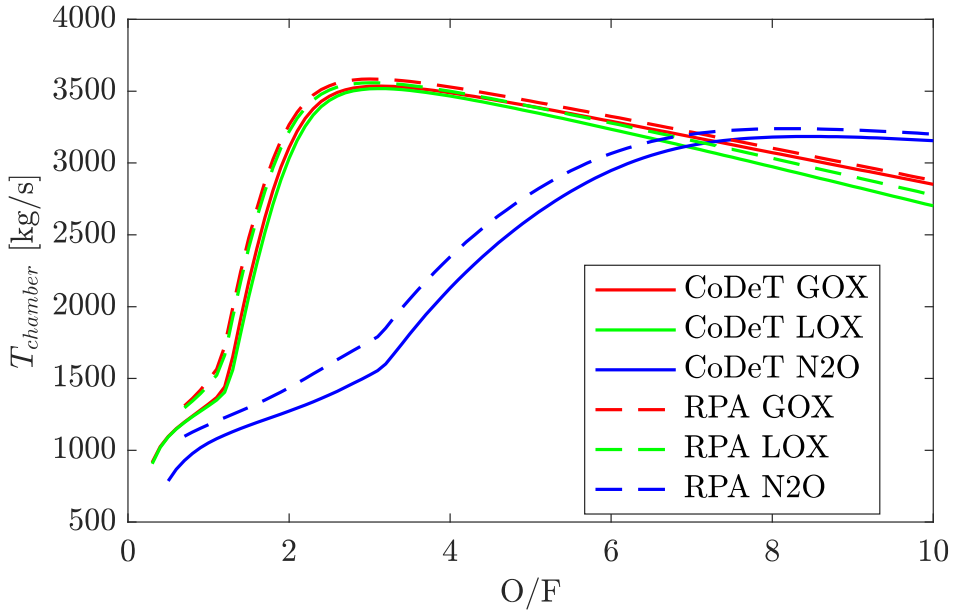


Figure 23 Validation of flame temperature data with RPA for the three engines (P_c = 30 bar, AR=2)

The comparison of chamber temperature values between RPA and CoDeT reveals discrepancies of 5% for LOX and GOX, and 7% for N_2O.

The overall shape and characteristics of the performance parameter curves generated by RPA and CoDeT demonstrate a close resemblance, indicating a general agreement between the two programs. However, a consistent trend is observed, with RPA consistently predicting higher performance values compared to CoDeT. This pattern is evident across specific impulse and characteristic velocity. The existence of these discrepancies between CoDeT and RPA results can be attributed to the utilization of different thermodynamic data in the respective software (CEA and RPA). These variations in thermodynamic data can lead to differences in the predicted combustion performance of the reactants, thereby giving rise to the observed discrepancies discussed earlier. While the precise reasons behind these discrepancies may be challenging to ascertain definitively, the distinct thermodynamic data employed by the two software packages emerges as a prominent contributing factor.

5.4.2 Combustion Analysis Validation

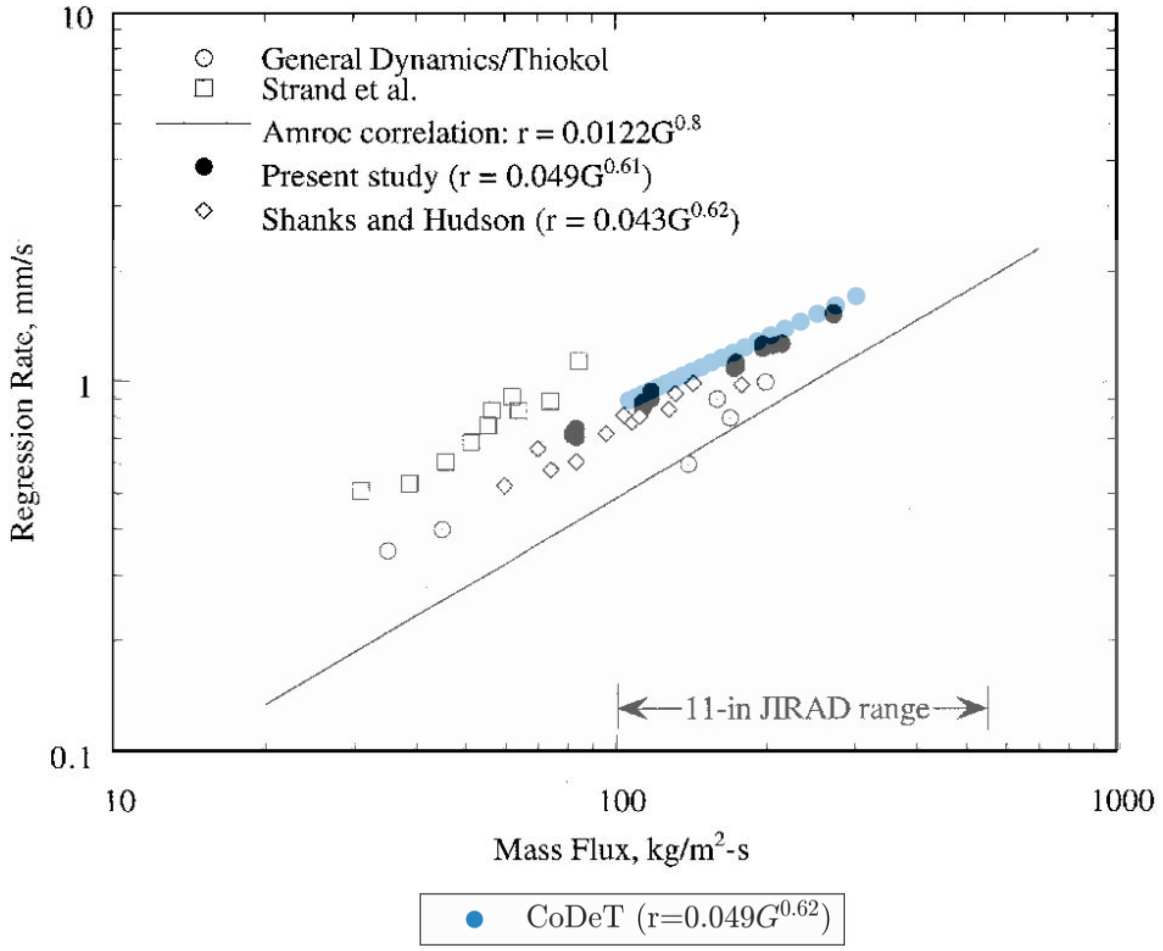


Figure 24 Validation of CoDeT GOX data with several regression rate studies [39, pp. 127, Fig. 3] [40] [41]

The temporal regression rate behavior of Paraffin/GOX is validated by the work of Chiaverini et al. [39]. In this study regression rate of HTPB/GOX is investigated experimentally. The test data attained by the study as well as data of various other studies [40] [41] are compared in Figure 24. The temporal regression rate and oxidizer mass flux data from the combustion analysis of Paraffin/GOX engine (as specified in the Table 11) are compared with the data from the mentioned studies. Note that the frequency of the CoDeT data points is reduced to 20 Hz for better visualization. The experimental data of Chiaverini et al., which uses almost the same values for a and n, matches very well with CoDeT regression rate data being slightly higher.

The extent of O/F shift and port diameter enlargement of oxygen hybrids during combustion is validated through the work of *Karabeyoglu et al.* titled *"O/F Shift" in Hybrid Rockets* [38]. In this work, the following analytical formula for predicting the final O/F ratio is derived and the analytical results are numerically confirmed.

$$(O/F)_{final} = (O/F)_{initial} R_f^{2n-1} \quad 4.1$$

Equation 4.1 [38] is used to validate the results obtained from the combustion analysis for the engines GOX and LOX. The parameter R_f represents the ratio of final to initial port diameter. By applying the formula, the final O/F ratios of the GOX and LOX engines at the end of the 10-second burn time are calculated and compared with the final O/F ratios calculated by CoDeT. The results from the analytical formula perfectly match the final O/F ratios calculated by CoDeT. This agreement suggests that the modeling of regression rate, port expansion, and O/F ratio changes during combustion has been accurately captured in the CoDeT combustion analysis.

5.4.3 Validation with ZEpHyR engine

The following validation of CoDeT is based on the development and subsequent testing of ZARM Experimental Hybrid Rocket (ZEpHyR) main engine. The theoretical design and construction of the full-scale engineering model (ZEpHyR EM-Engine) is detailed in the work of *Peter Rickmers et al.* titled *ZARM Experimental Hybrid Rocket: A contribution to the DLR STERN Project by the University of Bremen* [42]. The extensive testing of EM and development process of the flight version of the ZEpHyR main engine is presented in the work of *Peter Rickmers et al.* titled *ZEpHyR - ZARM Experimental Hybrid Rocket: Results of the Propulsion System Tests and Flight of a Small LOX/Paraffin Powered Sounding Rocket* [43].

The EM of the HRE and the subsequent flight version share identical internal geometry and performance output, including thrust, mass flow rate, pressure, and temperature. The EM has undergone extensive testing, as has the flight version, ensuring that both versions exhibit similar operational characteristics. Due to the well-tested nature of the EM and the flight version, their theoretical design parameters can be employed for validation purposes in this section of the study.

With the theoretical design parameters in Table 13, ZEpHyR EM-Engine is modeled using CoDeT and the results are presented in Table 14.

CoDeT input variable	
Oxidizer	LOX
Fuel	$C_{32}H_{66}$
Additive	-
Chamber pressure [bar]	35
Area ratio	5.14
Ambient pressure [atm]	1
Thrust output [N]	1900
Combustion efficiency	0.98
Initial oxidizer flux [$kg/(s\ m^2)$]	400
Burn time [s]	30
Time increment [s]	0.01
a	0.12
n	0.62

Table 13 Input parameters for the construction of the ZEpHyR engine in CoDeT

Grain design	**ZEpHyR**	**CoDeT prediction**
Initial port diameter [mm]	40	41
Final port diameter [mm]	140	101
Grain length [mm]	650	770
Burn time averaged performance parameters		
Thrust [N]	1800	1795
Oxidizer flux [kg/(s*m2)]	-	138
O/F	1.8	2.75
Oxidizer mass flow rate [kg/(s)]	≈ 0.5	0.53
Fuel mass flow rate [kg/(s)]	-	0.19
Regression rate [mm/s]	-	1
I_{sp} [s]	270	258
Thrust coefficient	-	1.5
c* [m/s]	-	1696
Adiabatic flame temperature [K]	-	3487

Table 14 Comparison of CoDeT output with ZEpHyR main engine EM and test data ("-" signifies that the values are not provided by the mentioned research papers)

Overall, the comparison between CoDeT and ZEpHyR EM shows a generally good agreement. Predicting very close values for initial port diameter, average thrust, and specific impulse. A notable discrepancy is observed in the grain design, where CoDeT predicts a grain with a higher length-to-diameter (L/D) ratio. This discrepancy can be attributed to the O/F ratios at which the engines operate. ZEpHyR operates at a more fuel-rich mixture ratio, while the engine modeled in CoDeT starts with an optimal O/F ratio of 2.3 and gradually increases during the burn time, reaching an average of 2.75.

Since the oxidizer mass flow rates in both engines are nearly the same, a lower O/F ratio in ZEpHyR results in a higher fuel mass flow rate. Considering that the burn time of both engines

are the same, this leads to ZEpHyR using more fuel, thus necessitating a larger grain and overall greater fuel mass.

6 Conclusion and Future Work

6.1 Summary of the Key Findings

This study developed CoDeT, a computational tool for HRE design and performance prediction. To show the tools capabilities, a test case of 1kN engine with 30 bar chamber pressure is designed and its performance using three different oxidizers is predicted. Based on the results of CoDeT, hybrid propulsion system design parameters are investigated and implications to the real-world application of the designed engines are discussed in detail.

Results obtained from CoDeT are compared with another preliminary design and performance analysis tool, namely RPA. The comparison revealed a generally good agreement. However, some discrepancies were observed in specific impulse and characteristic velocity values, particularly for N_2O. These discrepancies could be attributed to differences in thermodynamic data used in the models and warrant further investigation to the implications.

Combustion modeling of paraffin with oxygen is validated with experimental data found in the literature. Extent of O/F shift predicted in oxygen hybrids is confirmed with the analytical formula derived by Karabeyoglu et al. Furthermore, using CoDeT ZEpHyR EM is modeled and the results of test data and design parameters are compared. Here the validation was not completely satisfactory as the engine modeled by CoDeT operated at a different mixture-ratio than ZEpHyR EM. Consequently, the results exhibited some differences in grain design.

6.2 Limitations and Potential Areas for Improvement

As presented in this study, CoDeT offers the user a wide range of capabilities regarding the design of HREs and the validation of the results indicate overall an accurate representation of HRE performance. With some code improvements and further validation, the modeling capabilities of CoDeT can be significantly enhanced, leading to a more robust and reliable tool for HRE design.

Future research should focus on the validation of CoDeT predictions through comprehensive engine tests. Conducting real-world engine tests would yield essential data to verify the accuracy and reliability of CoDeT's performance predictions. In these tests, employing a highly instrumented EM would provide CoDeT with crucial engine parameters. Consequently, the analysis of the results would enable the necessary adjustments to be made to the code, making it more applicable and specific to the developed engine.

One objective of the engine tests should be to determine the regression rate behavior of the used propellant combination. This is done so by exploring a range of oxidizer flux values and

calculating the space-time averaged regression rate for each value. For these calculations, a series of parameters should be determined during the experimentation.

With the measured final port diameter, the average regression rate of a single firing test can be determined, additionally enabling the calculation of fuel mass flow rate. This method of calculation can deliver very accurate results. Oxidizer flux is determined by averaging the oxidizer mass flow rate with the port diameter. To ensure precise measurements of the oxidizer mass flow rate, the selection of an appropriate measurement method is essential. Among the available methods, using a coriolis or turbine flow meter is recommended as they provide continuous and accurate measurements of the oxidizer mass flow rate throughout the burn time. Under budgetary constraints it can also be measured by the weight loss of the oxidizer tank. However, this method comes at a cost of accuracy.

Upon conducting a series of burn tests, a mathematical correlation between the oxidizer flux and regression rate is established using the regression rate law. The correlation is subsequently implemented in CoDeT by configuring the parameters a and n, thus ensuring precise modeling of the temporal regression rate change for the specific propellant.

A limitation of CoDeT lies in the validation of injector design due to the scarcity of extensive test data available in the open literature for HREs. As a result, this section of CoDeT could not be adequately validated. To address this issue, cold-flow tests should be conducted to verify if the calculated mass flow rate by CoDeT aligns with the actual achieved values. Additionally, determining the required pressure drop value for sufficient injection and combustion stability is essential to further enhance the validation process and ensure the accuracy of CoDeT's predictions for injector design.

During firing tests, it is crucial to perform pressure and temperature measurements at various parts of the propulsion system such as at oxidizer tank, injector upstream and the chamber pressure. These measurements serve as valuable data to validate the accuracy of the thermodynamic conditions predicted during injector design and combustion modeling.

In conclusion, comprehensive engine tests and precise measurements are vital for further validation and improvement of CoDeT. By addressing these areas of concern, CoDeT can be refined into a more accurate and comprehensive tool for HRE design and performance analysis, offering valuable insights for future propulsion research and development.

7 References

[1] A. Okninski, W. Kopacz, D. Kaniewski and K. Sobczak, "Hybrid rocket propulsion technology for space transportation revisited - propellant solutions and challenges," *FirePhysChem,* 2021.

[2] E. Q. Granado, G. Pelenghi, J. Hijlkema, J. Anthoine and J.-Y. Lestrade, "A new System Design Tool for a Hybrid Rocket Engine Application," in *73rd International Astronautical Congress (IAC 2022)*, Paris, France, 2022.

[3] A. Ponomarenko, "RPA – Tool for Rocket Propulsion Analysis," in *Space Propulsion Conference*, Germany, 2014.

[4] M. J. Chiaverini and K. K. Kuo, Fundamentals of hybrid rocket combustion and propulsion, Reston, Va.: American Inst. of Aeronautics and Astronautics, 2007.

[5] N. Davydenko, R. Gollender, A. Gubertov, V. Mironov and N. Volkov, "Hybrid rocket engines: The benefits and prospects," *Aerospace Science and Technology,* p. 1, 2006.

[6] A. Karabeyoglu, "AA 284a/MECH 427 Advanced Rocket Propulsion Lecture 1," Istanbul, 2013.

[7] A. Karabeyoglu, "Performance Additives for Hybrid Rockets," in *Chemical Rocket Propulsion: A Comprehensive Survey of Energetic Materials*, Springer International Publishing Switzerland 2017, 2017, pp. 139-143.

[8] M. A. Karabeyoglu, D. Altman and B. J. Cantwell, "Combustion of Liquefying Hybrid Propellants: Part 1, General Theory," *JOURNAL OF PROPULSION AND POWER Vol. 18, No. 3* , pp. 3-10, 2002.

[9] W. Yin, S.-q. Hu, X.-l. Liu, Y. Zhang and L.-l. Liu, "Droplet entrainment and its role in the combustion of HTPB/paraffin fuels," *Acta Astronautica*, vol. 204, no. March, pp. 107-115, 2023 .

[10] A. Karabeyoglu, "AA 284a/MECH 427 Advanced Rocket Propulsion Hybrid Rocket Propulsion Lecture 8 Fundamentals," Stanford University&Koc University, Istanbul, 2013.

[11] G. P. Sutton and O. Biblarz, Rocket Propulsion Elements, Wiley-lnterscience, 2001.

[12] A. Karabeyoglu, AA 284a/MECH 427 Advanced Rocket Propulsion Lecture 2, 2013.

[13] D. Martinez-Sanchis, A. Sternin, T. Santese and O. J. Haidn, "The role of turbulence in the characteristic velocity and length of rocket combustors," *Aerospace Science and Technology*, vol. 134, 2023.

[14] V. Babu, Fundamentals of Propulsion, Springer, 2022.

[15] K. M. P. P Bose, "Motor, Analysis of Thrust Coefficient in a Rocket," *International Journal of Engineering and Advanced Technology,* vol. 1, no. 3, 2012.

[16] D. Mishra, Fundamentals of Rocket Propulsion, Florida: Taylor & Francis Group, 2017.

[17] G. Marxman and M. Gilbert, "Turbulent boundary layer combustion in the hybrid rocket," *Symposium (International) on Combustion,* vol. 9, no. 1, pp. 371-383, 1963.

[18] G. Zilliac and A. Karabeyoglu, "Hybrid Rocket Fuel Regression Rate Data and Modeling," in *42nd AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit* , Sacramento, California, 2006.

[19] A. Karabeyoglu, AA 284a/MECH 427 Advanced Rocket Propulsion Lecture 10, Department of Aeronautics and Astronautics Stanford University, 2013.

[20] A. Karabeyoglu, AA 284a/MECH 427 Lecture 9 Liquefying Fuels, Stanford University, 2013.

[21] Z. Wang, X. Lin, F. Li, J. Peng, Y. Liu, Z. Zhang, S. Fang and X. Yu, "Determining the time-resolved mass flow rates of hybrid rocket fuels using laser absorption spectroscopy," *Acta Astronautica,* vol. 188, pp. 110-120, 2021.

[22] C. Carmicino and A. Sorge, "Performance comparison between two different injector configurations in a hybrid rocket," *Aerosp. Sci. Technol. ,* p. 61–67. , 2007.

[23] C. Carmicino and A. Sorge, "Role of injection in hybrid rockets regression rate behaviour," *J. Propul. Power,* vol. 21, p. 606–612, 2005.

[24] B. S. Waxman, *An Investigation of Injectors for High Vapor Pressure Propellants in Hybrid Rockets A Dissertation Submitted to the Department of Aeronautics & Astronautics and the Committee on Graduate Studies of Stanford University for the Degree of Doctor of Philosop*, Stanford University, 2014.

[25] M. Bouziane, A. Bertoldi, P. Hendrick and M. Lefebvre, "Experimental investigation of the axial oxidizer injectors geometry on a 1-kN paraffin-fueled hybrid rocket motor," *FirePhysChem*, p. 231–243, 2021.

[26] Y. A. Çengel and M. A. Boles, *Thermodynamics: An Engineering Approach 8th Edition*, McGraw-Hill Education, 2015.

[27] J. Dyer, E. Doran, Z. Dunn and K. Lohner, "Modeling Feed System Flow Physics for Self-Pressurizing Propellants," in *43rd AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit* , Cincinnati, OH, 2007.

[28] C. E. Brennen, *Fundamentals of Multiphase Flows*, California: Cambridge University Press , 2005.

[29] J. P. d. l. B. M. B. B. Genevievea, "A Computational Tool for Predicting Hybrid Rocket Motor Performance," *R & D Journal of the South African Institution of Mechanical Engineering*, vol. 33, pp. 56-65, 2017,.

[30] P. M. Zolla, M. T. Migliorino, D. Bianchi, F. Nasuti, R. C. Pellegrini and E. Cavallini, "A Computational Tool for the Design of Hybrid Rockets," *Aerotec. Missili Spaz.*, vol. 100, p. 253–262, 2021.

[31] O. Božić, D. Porrmann, D. Lancelle and A. Hartwig, "Program AHRES: Assessing Features and Current Limitations of Hybrid Rocket Propulsion," in *63rd International Astronautical Congress*, Naples, 2012.

[32] S. Gordon and B. J. Mcbride, *Computer program for calculation of complex chemical equilibrium compositions and applications Part 1: Analysis,* Cleveland: NASA , 1994.

[33] G. R. Center, "NASA Technology Transfer Program," NASA, [Online]. Available: https://software.nasa.gov/software/LEW-17687-1. [Accessed 2023].

[34] S. Gordon and B. J. McBride, *Computer Program for Calculation of Complex Chemical Equilibrium Compositions and Applications II. Users Manual and Program Description,* Cleveland,OH: NASA , 1996.

[35] C. Foster, *CEAwin: NASA CEA interface,* not published.

[36] N. I. o. S. a. Technology, "NIST Chemistry Webbook SRD 69," NIST, [Online]. Available: https://webbook.nist.gov/chemistry/fluid/. [Accessed 07 2023].

[37] B. Kahraman, H. Karakas, B. N. Eren and A. Karabeyoglu, "Erosion Rate Investigation of Various Nozzle Materials in Hybrid Rocket Motors," in *AIAA Propulsion and Energy 2020 Forum*, VIRTUAL EVENT, August 24-28, 2020.

[38] A. Karabeyoglu, E. Toson and B. Evans, "“O/F Shift” in Hybrid Rockets," in *50th AIAA/ASME/SAE/ASEE Joint Propulsion Conference*, Cleveland, OH, July 28-30, 2014.

[39] M. J. Chiaverini, N. Serin, D. K. Johnson, Y.-C. Lu, K. K. Kuo and G. A. Risha, "Regression Rate Behavior of Hybrid Rocket Solid Fuels," *JOURNAL OF PROPULSION AND POWER*, vol. 16, no. 1, pp. 125-132, 2000.

[40] R. Shanks and M. Hudson, "The Design andControl of a Labscale Hybrid Rocket Facility for Spectroscopy Studies," in *Joint Propulsion Conference*, Indianapolis, 1994.

[41] L. Strand, M. D. Jones, R. L. Ray and N. S. Cohen, "Characterization of Hybrid Rocket Internal Heat Flux and HTPB Fuel Pyrolysis," in *AIAA/ASME/SAE/ASEE Joint Propulsion Conference*, Indianapolis, 1994.

[42] P. Rickmers, M. Drinkewitz and F. Orlowski, "The propulsion system of the ZEpHyR - ZARM Experimental Hybrid Rocket: A contribution to the DLR STERN Project by the University of Bremen," in *5TH EUROPEAN CONFERENCE FOR AERONAUTICS AND SPACE SCIENCES (EUCASS)*, 2013.

[43] P. Rickmers, F. Ruhammer and T. Ganser, "ZEpHyR - ZARM Experimental Hybrid Rocket: Results of the Propulsion System Tests and Flight of a Small LOX/Paraffin Powered Sounding Rocket," in *7TH EUROPEAN CONFERENCE FOR AERONAUTICS AND SPACE SCIENCES (EUCASS)*, 2017.

[44] A. Karabeyoglu, *AA 284a/MECH 427 Advanced Rocket Propulsion Lecture 4*, KOC University, 2013.